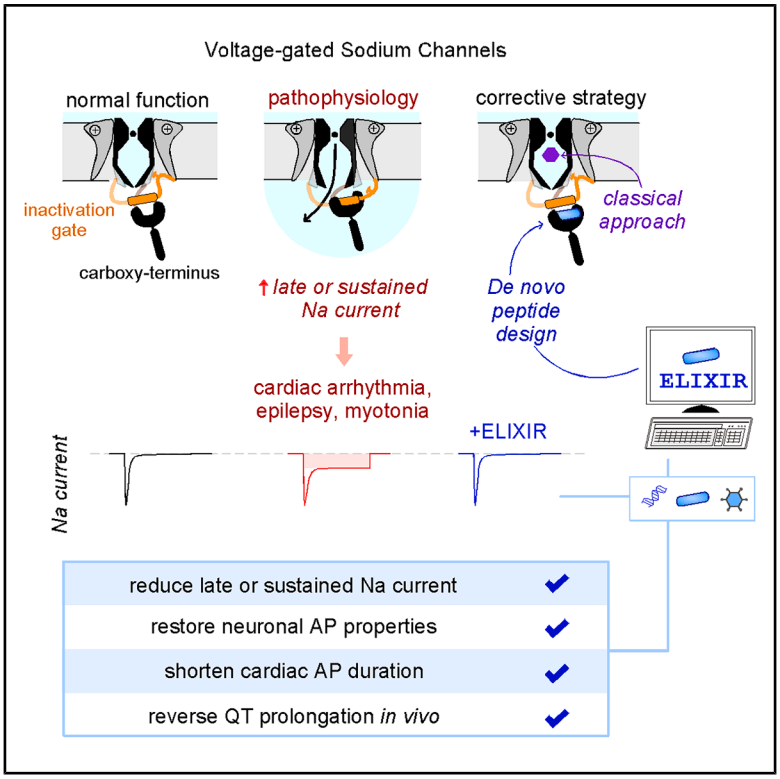
Cell

De novo design of a peptide modulator to reverse sodium channel dysfunction linked to cardiac arrhythmias and epilepsy

Graphical abstract



Highlights

- A *de novo* designed peptide, ELIXIR, reverses gain-of-function defects in Na channels
- ELIXIR demonstrates selectivity in targeting "pathogenic" late sodium current
- In neurons from an epilepsy-linked SCN8A model, ELIXIR reduces depolarization block
- ELIXIR shortens the cardiac action potential duration in multiple disease models

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In brief

De novo protein design was used to engineer a peptide that reverses gain-of-function deficits in voltage-gated sodium channels linked to cardiac arrhythmia and epilepsy.

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Article

De novo design of a peptide modulator to reverse sodium channel dysfunction linked to cardiac arrhythmias and epilepsy

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SUMMARY

Ion channels orchestrate electrical signaling in excitable cells. In nature, ion channel function is customized by modulatory proteins that have evolved to fulfill distinct physiological needs. Yet, engineering synthetic modulators that precisely tune ion channel function is challenging. One example involves the voltage-gated sodium (Na_V) channel that initiates the action potential and whose dysfunction amplifies the late/persistent sodium current (I_{NaL}), a commonality that underlies various human diseases, including cardiac arrhythmias and epilepsy. Here, using a computational protein design platform, we engineered a *de novo* peptide modulator, engineered late-current inhibitor X by inactivation-gate release (ELIXIR), that binds Na_V channels with submicromolar affinity. Functional analysis revealed unexpected selectivity in inhibiting "pathogenic" I_{NaL} and confirmed its effectiveness in reversing Na_V dysfunction linked to both cardiac arrhythmias and epilepsy in cellular and murine models. These findings exemplify the efficacy of *de novo* protein design for engineering synthetic ion channel modulators and set the stage for the rational design of future therapeutic approaches.

INTRODUCTION

Ion channels are pore-forming proteins that serve as nature's transistor.[1] Unlike their silicon counterparts, ion channel activity is dynamic and tightly regulated, allowing cells to adapt their electrical properties to match the complex requirements of life. Indeed, ion channel dysfunction underlies life-threatening human diseases, including cardiac arrhythmias,[2–4] epilepsy,[5] myotonia,[6] migraines,[7] and intellectual disability.[8] For many voltage-gated ion channels, pathophysiological changes in channel function result in altered channel dynamics that disrupt overall electrical signaling. As such, strategies to precisely tune ion channel function are highly desired, yet current approaches, including small-molecule pharmacology, are limited and often have off-target effects.[9,10] In nature, the functional plasticity of ion channels, a defining feature of these proteins, is often attained through association with diverse auxiliary subunits or regulatory proteins that have evolved to fine-tune specific channel properties.[11] Structural studies over the past several decades have furnished an unprecedented atomistic view of such interactions and shed light onto mechanisms of regulation.[12–15] Yet, rational engineering of *de novo* or synthetic ion channel modulators with high specificity has remained intractable.

An example of this deficiency involves the family of voltage-gated sodium channels ($Na_V1.1$–1.9), which are responsible for the generation and propagation of action potentials (APs) in excitable cells. A hallmark of Na_V channel function is the fast activation and inactivation that drives the rapid depolarization of many excitable cells, including neurons and muscle cells. Disruption of Na_V channel function is linked to a broad spectrum of clinical phenotypes, including cardiac arrhythmias, congenital myotonia, epilepsy, neurodevelopmental disorders, and painful neuropathies. Despite their distinct etiology, many of these disorders are linked to deficits in channel inactivation that result in sustained Na^+ influx, referred to as either late or persistent Na^+ current (I_{NaL}). Two prominent examples involve the *SCN5A* gene, which encodes $Na_V1.5$ in the heart, and the *SCN8A* gene, which encodes $Na_V1.6$ in the nervous system.

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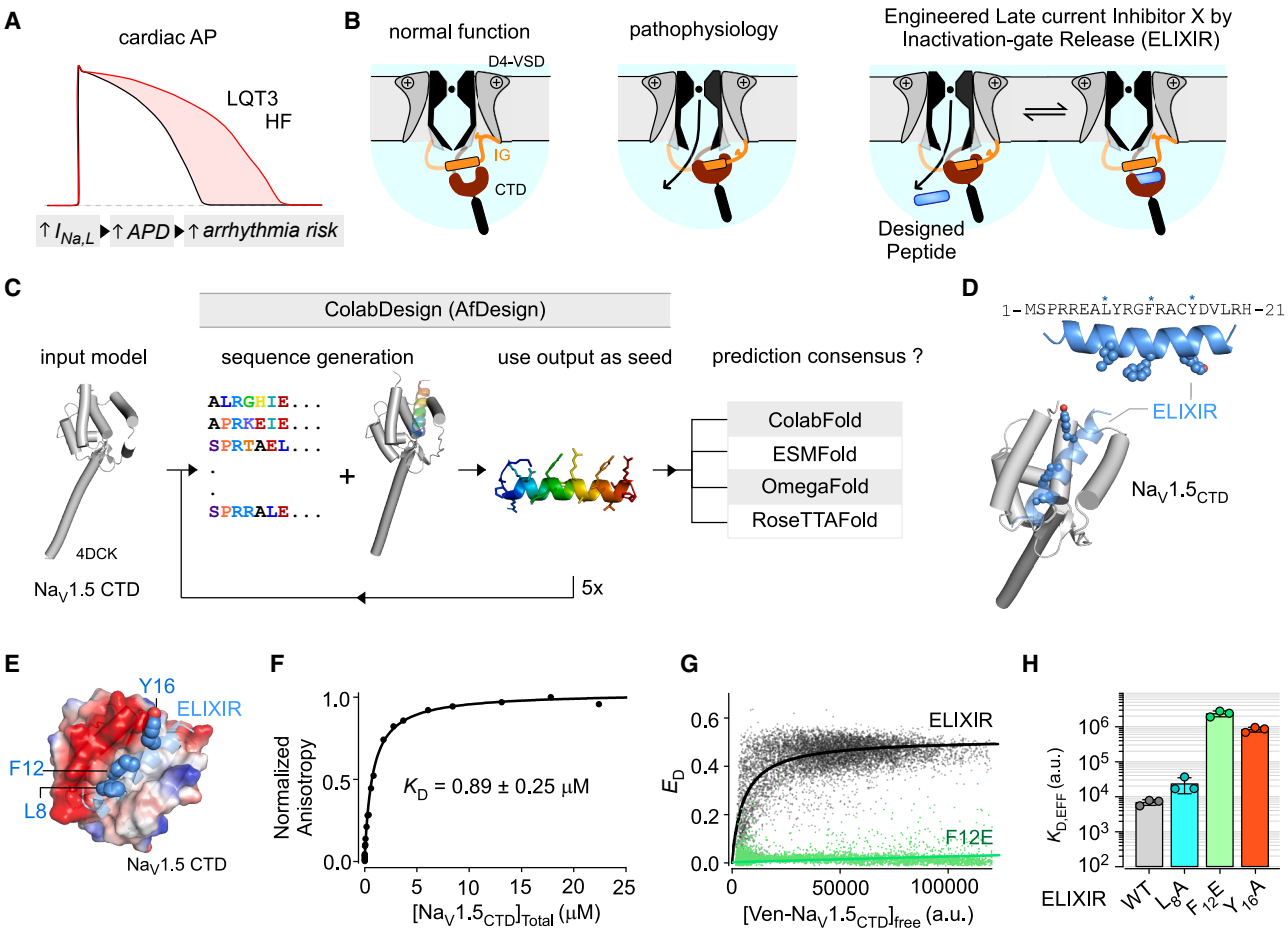


Figure 1. *De novo* design of a peptide modulator of $Na_V1.5$

(A) Increased I_{NaL} prolongs AP duration and increases the risk of cardiac arrhythmias.

(B) Conceptual scheme for design of an I_{NaL} inhibitor. Left, following activation, the IG binds a pore-proximal site to inactivate $Na_V1.5$ (black/gray; IG, orange; EFL, dark red). Middle, in pathophysiology, IG remains associated with CTD. Competitive displacement of IG from EFL by ELIXIR inhibits I_{NaL}.

(C) Schematic shows the peptide design framework. The initial coordinates of the $Na_V1.5$ CTD (PDB: 4DCK) served as input into AfDesign to hallucinate a peptide binder.

(D) Top, helical model of ELIXIR along with residues L8, F12, and Y16 highlighted. Bottom, ColabFold predicted model of the $Na_V1.5$ CTD-ELIXIR complex.

(E) ColabFold prediction of the EFL region of $Na_V1.5$ CTD bound to ELIXIR. Vacuum electrostatic surface (positive: blue, neutral: white; negative: red).

(F) Equilibrium titration of ELIXIR with the $Na_V1.5_{CTD}$ shows saturable binding. K_D shown as mean ± SD from n = 3 titrations.

(G) FRET-binding curves with Venus-tagged $Na_V1.5_{CTD}$ and cerulean-tagged ELIXIR (black) or ELIXIR F12E mutant (green). a.u., arbitrary units. The solid line depicts the fit of a 1:1-binding isotherm.

(H) Bar graph summary of $K_{d,EFF}$, a relative dissociation constant for ELIXIR-$Na_V1.5_{CTD}$ interaction from FRET two-hybrid experiments.

See also Figures S1 and S2.

Specifically, gain-of-function (GoF) human variants in $Na_V1.5$ that upregulate I_{NaL} are linked to arrhythmogenic long-QT syndrome 3 (LQT3)[16] and atrial fibrillation.[2] Similarly, pathological remodeling associated with heart failure (HF)[3] or following myocardial ischemia[17] has been shown to modify $Na_V1.5$ function and elevate I_{NaL}, resulting in an increased risk of cardiac arrhythmias. Increased I_{NaL} causes a proarrhythmic prolongation of the AP[4,18,19] (Figure 1A) and disrupts Na^+ and Ca^{2+} homeostasis, which sets off a vicious and detrimental cycle involving Ca^{2+}/calmodulin (CaM)-dependent kinase II mutant (CaMKII), reactive oxygen species (ROS), and ryanodine receptor (RyR).[20] In the nervous system, GoF $Na_V1.6$ variants are linked to severe developmental epilepsies and sudden unexpected death in epilepsy (SUDEP).[21] These variants disrupt channel inactivation and alter neuronal AP properties to ultimately drive neuronal network hyperexcitability.

Beyond cardiac and epileptic disorders, increased I_{NaL} through related channels expressed in skeletal muscle or the peripheral nervous system is associated with congenital myotonia[6] and pain disorders,[22] pointing to the potential utility of I_{NaL} inhibitors. Despite advances in the understanding of pathophysiological mechanisms and the availability of high-resolution structures,[23,24] devising effective strategies to reverse changes in Na_V function has been limited. The prevailing approach has

been small-molecule I_{NaL} blockers such as ranolazine, which show some promise. However, these drugs target a common antiarrhythmic drug (AAD)-binding site in the transmembrane region, which may variably inhibit peak Na^+ current or block other ion channels.[9] In addition, these small molecules may also have reduced affinity for disease-linked mutant channels, limiting their potential utility. Recently, certain auxiliary proteins, such as the intracellular fibroblast growth factor homologous factors (iFGF/FHFs), have been shown to inhibit I_{NaL} with minimal effects on peak Na current.[25,26] Additionally, peptides derived from iFGF/FHF have been shown to be sufficient to recapitulate these effects.[26] Even still, despite optimization, these peptides are comparatively large and show reduced efficacy for certain channel variants. To this end, major advances in machine-learning-based computational protein design have increased the accuracy of *de novo* protein design while reducing computational time.[27–29] These emerging methods may provide an alternative strategy to rationally design peptide modulators that precisely modify Na_V function. If substantiated, these approaches may be generalized to engineer a toolkit of synthetic ion channel regulators.

With cardiac $Na_V1.5$ as a template, we used *de novo* protein design to generate a peptide modulator that selectively inhibits pathogenic I_{NaL} by enhancing a native mechanism of channel inactivation. To do so, we implemented a protocol for peptide hallucination within the ColabDesign framework[30,31] and leveraged this to generate a short 21-amino acid (aa) peptide targeting the $Na_V1.5$ carboxy-terminal domain (CTD), an intracellular channel domain that serves as a regulatory hub. *In vitro*-binding analysis confirmed the interaction of the candidate peptide with the $Na_V1.5$ CTD. In-depth electrophysiological analysis confirmed the functionality of the *de novo* designed peptide and revealed an unexpected selectivity of the designed peptide for pathogenic I_{NaL} of multiple members of the Na_V family. We further demonstrate the effectiveness of this peptide in both neuronal and cardiac disease models. Our findings highlight the utility of *de novo* protein design in the development of synthetic ion channel modulators that tune specific aspects of ion channel function in physiology and pathophysiology.

RESULTS

A rational approach to developing an I_{NaL} inhibitor

Fast inactivation of Na_V channels is a multistep process involving distinct channel elements. Following depolarization, movement of voltage-sensing domains (VSDs) I–III couples to pore opening, while that of domain IV VSD initiates conformational changes that result in inactivation.[32–34] Of particular importance for the latter is the inactivation gate (IG, linker between domains III and IV), which contains the isoleucine-phenylalanine-methionine (IFM) motif that binds to a site near the channel pore to halt Na^+ influx[35,36] (Figure 1B). In addition to this pore-proximal site, the IG also binds an acidic EF-hand-like region (EFL) in the channel CTD,[37–40] suggesting that the IG may first need to be released from this EFL site to interact with the pore-proximal site and halt Na^+ influx. In cardiac pathologies, this process is disrupted, resulting in decreased $Na_V1.5$ inactivation and increased I_{NaL},[4,19] suggesting reduced occupancy of the pore-proximal site by the IG. Given this, we reasoned that dislodging the IG from its CTD interaction site would promote Na_V inactivation, and in so doing, reduce I_{NaL} and AP duration (APD; Figure 1B). We used ColabDesign[31] to generate a peptide targeting the EFL of the $Na_V1.5$ CTD as depicted in Figure 1C. This process yielded a small (21 aa) peptide with features reminiscent of the $Na_V1.5$ IG (hydrophobic surface flanked by basic residues), consistent with it being able to bind the hydrophobic cleft in the EFL of the $Na_V1.5$ CTD (Figures 1D and 1E). Given its putative mechanism, we termed this peptide *engineered late-current inhibitor X by inactivation-gate release* (ELIXIR). As an initial filter, we compared models of the ELIXIR-$Na_V1.5$ CTD (aa 1,773–1,940, $Na_V1.5_{CTD}$) complex generated with several structure prediction algorithms[27,41–43] (Figure S1A). In all models, the $Na_V1.5$ EFL and IQ motif adopted a nearly identical architecture (backbone root-mean-square deviation [RMSD] $\sim$ 1.58 Å) as in crystallographic structures[44–46] (Figure S1A), lending credence to the predicted models. In each, ELIXIR bound to the $Na_V1.5$ EFL in an identical orientation. The interface between ELIXIR and $Na_V1.5$ EFL was highly similar across the models, with peptide residues L8, F12, and Y16 making close contacts to residues in the hydrophobic cleft of the EFL (Figures S1A–S1I). Beyond this, ColabFold models of the $Na_V1.5$ CTD-CaM-$Na_V1.5$ IG complex in the absence and presence of ELIXIR show ELIXIR bound to the EFL in place of the IG (Figure S1J), suggesting it may displace the IG from this site. Comparison of our designed peptide sequence to that of the $Na_V1.5$ IG revealed only $\sim$24% identity (Figure S2A), indicating that our designed sequence was not merely recapitulating the native ligand of the $Na_V1.5$ EFL. Indeed, when searched against the BLAST database,[47] the closest homolog to ELIXIR was a 50S ribosome of the algae *Nannochloropsis gaditana* ($\sim$52% identity; Figure S2A). The closest mammalian homolog was DNA replication factor CDT1 of the Weddell seal (*Leptonychotes weddellii*; Figure S2A). A single conservative substitution was made to the sequence (C15S) to simplify *in vitro* applications of ELIXIR. Having confirmed that the ELIXIR sequence was both *de novo* and foreign to $Na_V1.5$, we examined whether it binds directly to the channel CTD. Equilibrium titrations of a fluorescein-tagged ELIXIR with the $Na_V1.5_{CTD}$, monitored through fluorescence anisotropy, showed saturable binding with a submicromolar affinity (K_D = 0.89 ± 0.25 μM; Figure 1F). To ensure that ELIXIR associates with the $Na_V1.5_{CTD}$ in the milieu of living cells, we undertook a flow-cytometry-based fluorescence resonance energy transfer (FRET) two-hybrid assay. Co-expression of cerulean-tagged ELIXIR, a FRET donor, and Venus-tagged $Na_V1.5_{CTD}$, a FRET acceptor, resulted in robust FRET consistent with this interaction in mammalian cells (Figure 1G). We sought to validate the interface between ELIXIR and $Na_V1.5$ CTD. Given the potential importance of ELIXIR residues L8, F12, and Y16 based on predicted models, we reasoned that the substitution of these aliphatic residues with alanine (L8 and Y16) or glutamate (F12) would disrupt the ELIXIR-$Na_V1.5$ CTD interaction. FRET two-hybrid analysis confirmed this expectation, with each mutant weakening binding to a different degree (Figures 1G, 1H, S2B, and S2C). Collectively, these results demonstrate that *de novo* designed ELIXIR peptide interacts with $Na_V1.5$ CTD and may be positioned to tune channel function.

ELIXIR decreases I_{NaL} of a channelopathic $Na_V1.5$ variant

Thus affirmed, we sought to determine the functional consequences of ELIXIR interaction on $Na_V1.5$ function. Whole-cell patch-clamp recordings demonstrated that ELIXIR co-expression did not significantly alter channel activation, inactivation, or recovery from inactivation of wild-type $Na_V1.5$ (Figures S2D–S2G). To probe whether ELIXIR inhibits pathological I_{NaL}, we conducted cell-attached multichannel recordings of HEK293 cells heterologously expressing either wild-type $Na_V1.5$ or a disease-linked variant that truncates the channel CTD (Δ1864). As the channel unitary current is larger than instrument noise, multichannel recordings allow simultaneous quantification of peak current and rare late channel openings.[25,26,48] We quantified I_{NaL} as $R_{persist}$, the ratio of average open probability (P_O) following 50 ms of depolarization normalized to the peak P_O. Wild-type $Na_V1.5$ exhibited a low baseline I_{NaL}, which was minimally perturbed by co-expression of ELIXIR (Figures 2A–2C and S3A). By contrast, the Δ1864 mutant showed greatly (∼40-fold) elevated I_{NaL}, consistent with its pathogenicity (Figures 2C–2E and S3B). Strikingly, ELIXIR co-expression evoked a >12-fold reduction in I_{NaL} of the Δ1864 channel (Figures 2C–2E and S3B). A similar degree of I_{NaL} inhibition was observed with the unmodified ELIXIR sequence that lacked the C15S substitution (Figures S3C–S3E). Native Na_V channel complexes may include auxiliary β-subunits, which modulate I_{NaL}.[49,50] ELIXIR minimally perturbed wild-type $Na_V1.5$ co-expressed with β1 and β4 (Figures S3F, S3H, and S3J). By comparison, expression of β1 and β4 with the Δ1864 mutant yielded a reduction in $I_{Na,L}$, which was further reduced with ELIXIR (Figures S3G, S3I, and S3J). Ranolazine (10 μM), a well-established small-molecule I_{NaL} inhibitor,[24] evoked an ∼4-fold reduction in I_{NaL} of $Na_V1.5$ Δ1864 (Figures 2C, 2F, and S4A).

Having confirmed its functionality, we investigated whether the change in I_{NaL} observed with ELIXIR results from its bona fide interaction with $Na_V1.5$ CTD. We first examined how mutations of key ELIXIR residues that weaken binding to the $Na_V1.5$ CTD alter functional inhibition of I_{NaL} (Figure 2G). We found that F12E strongly diminished I_{NaL} inhibition, while Y16A and L8A evoked partial effects (Figures 2H, 2I, and S4B–S4F). Quantitative analysis revealed a correlation between the level of I_{NaL} ($R_{persist}$) and the relative dissociation constant ($K_{D,EFF}$) determined by FRET analysis (Figure 2J), pointing to the importance of the ELIXIR-$Na_V1.5$ CTD interaction. Second, we reasoned that deletion of the ELIXIR-binding site (i.e., the EFL) would abolish I_{NaL} inhibition by ELIXIR. To test this, we considered the channelopathic $Na_V1.5$ Δ1810 variant, which encodes only the first helix of the EFL (Figure 2K). As with the Δ1864 mutant, $Na_V1.5$ Δ1810 showed a markedly increased basal I_{NaL} relative to the wild-type channel (Figures 2L, 2M, and S4G). However, unlike $Na_V1.5$ Δ1864, co-expression of ELIXIR had virtually no effect on I_{NaL} (Figures 2L, 2M, and S4G). Together, the reduced efficacy of ELIXIR mutants in inhibiting I_{NaL} and the inability of ELIXIR to inhibit I_{NaL} of $Na_V1.5$ Δ1810 suggest that ELIXIR interacts with holo-$Na_V1.5$ via an interface similar to that predicted in the models of the ELIXIR-$Na_V1.5_{CTD}$ complex.

ELIXIR is a general inhibitor of $Na_V1.5$ I_{NaL}

At the molecular level, a variety of perturbations to $Na_V1.5$ contribute to pathogenic increases in I_{NaL}. These include variants distributed throughout the channel sequence and post-translational modifications, such as phosphorylation, that are associated with cardiac pathologies, including HF.[51] Our results with $Na_V1.5$ Δ1864 suggest that ELIXIR is a potent inhibitor of I_{NaL}; thus, we sought to determine its generality in tuning I_{NaL} triggered by distinct mechanisms. Accordingly, we assessed the efficacy of ELIXIR in inhibiting I_{NaL} of several $Na_V1.5$ mutations located in distinct channel domains. These included (1) the ΔKPQ variant linked to LQT3, (2) a mutation (F1759A) in the classic AAD-binding pocket linked to atrial fibrillation,[2] and (3) a structure-guided mutation (IQ/AA) engineered to block binding of the auxiliary protein CaM (Figure 3A). These mutations were selected as they represent disparate structural mechanisms that can contribute to increased I_{NaL} in cardiac pathologies. Consistent with our previous results,[26] when expressed alone, each of these $Na_V1.5$ mutants displayed an elevated I_{NaL} (21-fold F1759A; 13-fold ΔKPQ; 10-fold IQ/AA) relative to the wild-type channel (Figures 3B, 3C, 3E, and S5A–S5D). For each $Na_V1.5$ mutant, co-expression of ELIXIR resulted in a statistically significant decrease in I_{NaL}, with reductions ranging from 12-fold (F1759A) to 3-fold (IQ/AA; Figures 3B, 3C, 3E, and S5A–S5D).

Post-translational modification of $Na_V1.5$ also contributes to elevated I_{NaL} and is associated with increased risk of arrhythmia in both HF[51,52] and myocardial ischemia.[53,54] To determine whether ELIXIR inhibits phosphorylation-dependent I_{NaL}, we co-expressed wild-type $Na_V1.5$ with the PKA catalytic subunit (PKA_{cat}) or a constitutively activated $CaMKII_{T286D}$. In both cases, we observed an increase in basal I_{NaL} (∼5-fold with PKA_{Cat} or ∼2-fold with $CaMKII_{T286D}$; Figures 3D, 3E, and S5E–S5G). Co-expression of ELIXIR sufficed to reverse these changes in I_{NaL} (Figures 3D, 3E, and S5E–S5G). The ability of ELIXIR to inhibit I_{NaL} stemming from a wide range of molecular alterations (i.e., mutations in distinct channel domains and post-translational modifications) indicates that it may be generalizable and further highlights its potential utility.

We next examined the dose-dependence of I_{NaL} inhibition by ELIXIR. To facilitate intracellular delivery, we synthesized ELIXIR as a peptide fused to the cell-penetrating region of the transduction domain of HIV-1 *trans* activator of transcription with an N-terminal fluorescein marker (ELIXIR-cpp). Incubation of ELIXIR-cpp for 4 h yielded a dose-dependent reduction in I_{NaL} with an IC_{50} of ∼330 ± 230 nM (95% confidence interval [CI]), similar to the *in vitro* affinity of ELIXIR for the $Na_V1.5_{CTD}$ (Figures 3F and S5H–S5K).

ELIXIR preferentially tunes pathological I_{NaL}

The nine Na_V isoforms (1.1–1.9) are known to have distinct inactivation properties that are critical for their physiological functions.[55] GoF variants in Na_V isoforms are linked to diverse clinical phenotypes, including epilepsy,[21] myotonia,[6] neurodevelopmental disorders,[56] and chronic pain[22] (Figure 4A). The high homology of the CTD (Figure 4B) suggests that ELIXIR may interact with each Na_V isoform. Consistent with this, in ColabFold[41]-generated models, ELIXIR was predicted to adopt a highly similar binding pose with the EFL of $Na_V1.1$–1.9 (Figures S6A–S6H).

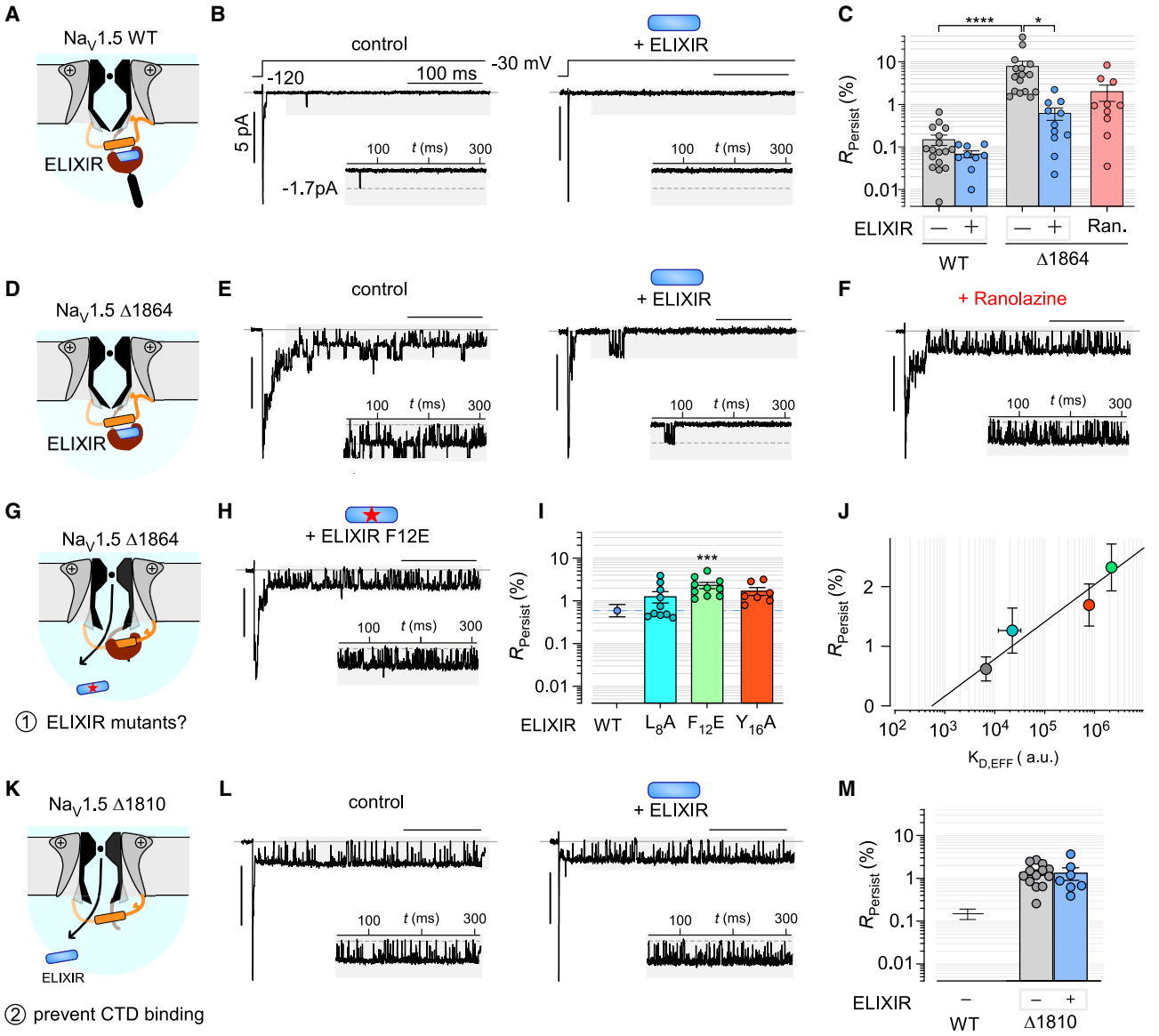


Figure 2. Functional inhibition of I_{NaL} by ELIXIR

(A) Schematic shows ELIXIR interaction with wild-type (WT) Na$_V$1.5.

(B) Exemplar cell-attached multichannel recordings of WT Na$_V$1.5 in HEK293 cells both without (left) and with ELIXIR (right). Channel openings are downward deflections from the zero line (gray). Inset, the late phase enlarged for visualization.

(C) Bar graph summary shows changes in I_{NaL} quantified as $R_{persist}$. WT Na$_V$1.5 without (n = 17 cells; 1,069 sweeps) and with ELIXIR (n = 9 cells; 924 sweeps). Na$_V$1.5 Δ1864 without ELIXIR (n = 17 cells; 907 sweeps), with ELIXIR (n = 11 cells; 948 sweeps), or with 10 μM ranolazine (n = 10 cells; 848 sweeps). Each bar, mean ± SEM. ****p < 0.001 compared with WT, *p = 0.0147 compared with Na$_V$1.5 Δ1864 + ELIXIR by Kruskal-Wallis test followed by Dunn's multiple comparison test.

(D–F) I_{NaL} of Na$_V$1.5 Δ1864 with ELIXIR (E) or ranolazine (F). Format as in (A)–(C).

(G) Schematic shows a weak association of the ELIXIR mutant F12E to Na$_V$1.5 Δ1864 mutant.

(H) Exemplar recordings of Na$_V$1.5 Δ1864 mutant with ELIXIR F12E mutant.

(I) Bar graph summary of changes in $R_{persist}$ of Na$_V$1.5 Δ1864 with ELIXIR L8A (n = 10 cells; 852 sweeps), ELIXIR F12E (n = 10 cells; 885 sweeps), or ELIXIR Y16A (n = 7 cells; 671 sweeps) mutants. Gray, $R_{persist}$ of Na$_V$1.5 Δ1864 alone; blue, with ELIXIR. Each bar, mean ± SEM. ***p = 0.0008 (+ELIXIR F12E), compared with Na$_V$1.5 Δ1864 + ELIXIR by Kruskal-Wallis test followed by Dunn's multiple comparison test.

(J) Dot plot shows relationship between $R_{persist}$ of Na$_V$1.5 Δ1864 (mean ± SEM) with WT or mutant ELIXIR from (I) and corresponding $K_{D,EFF}$ reproduced from Figure 1H. Black, trend-line.

(K) Schematic shows Na$_V$1.5 Δ1810 mutant that lacks the ELIXIR-interacting CTD interface.

(L) Exemplar recordings of Na$_V$1.5 Δ1810 mutant with ELIXIR.

(M) Bar graph summary of changes in $R_{persist}$ of Na$_V$1.5 Δ1810 without (n = 13 cells, 694 sweeps) or with ELIXIR (n = 7 cells, 593 sweeps).

See also Figures S3 and S4.

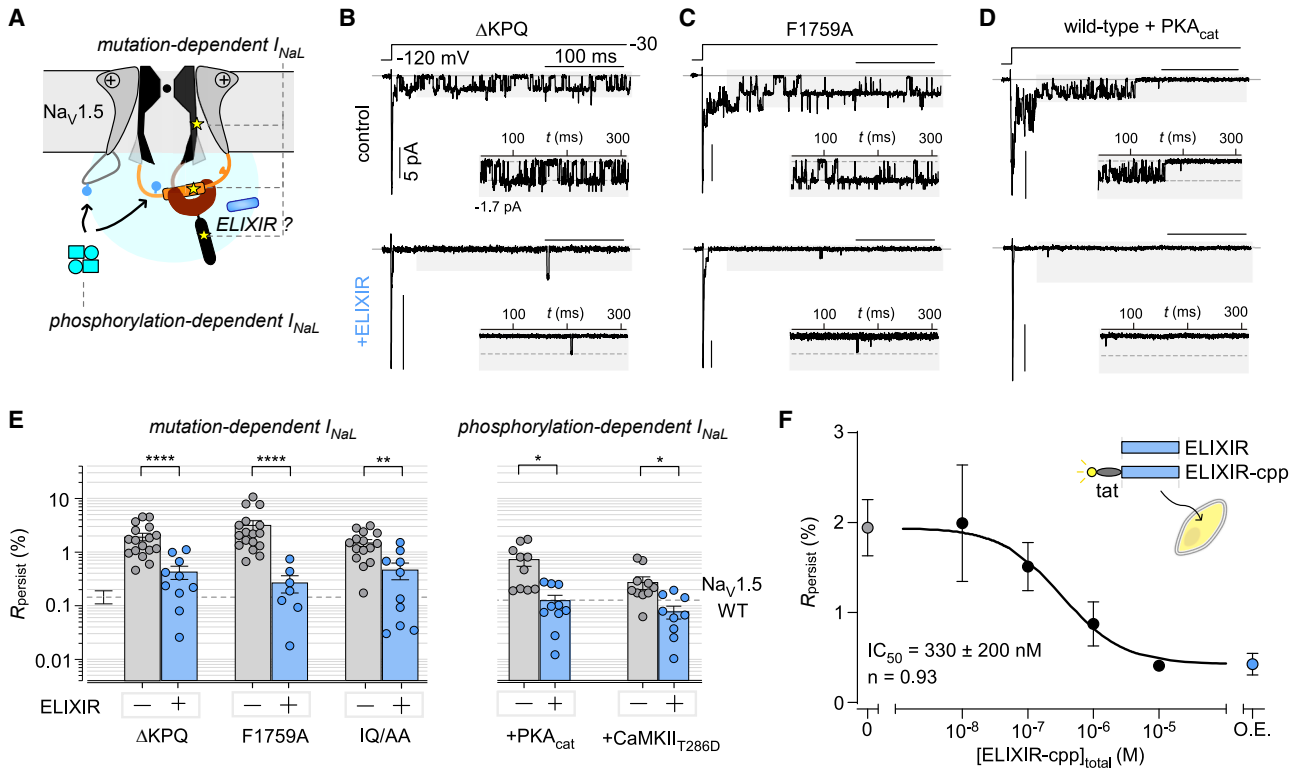


Figure 3. Generality of late Na^+ current inhibition by ELIXIR

(A) Schematic showing that either mutations (stars) in or phosphorylation (circles) of $Na_V1.5$ can lead to increased I_{NaL}.

(B–D) Exemplar recordings of $Na_V1.5$ ΔKPQ (B), $Na_V1.5$ F1759A (C), and WT $Na_V1.5$ with PKA_{Cat} (D) at baseline (top) and with ELIXIR (bottom).

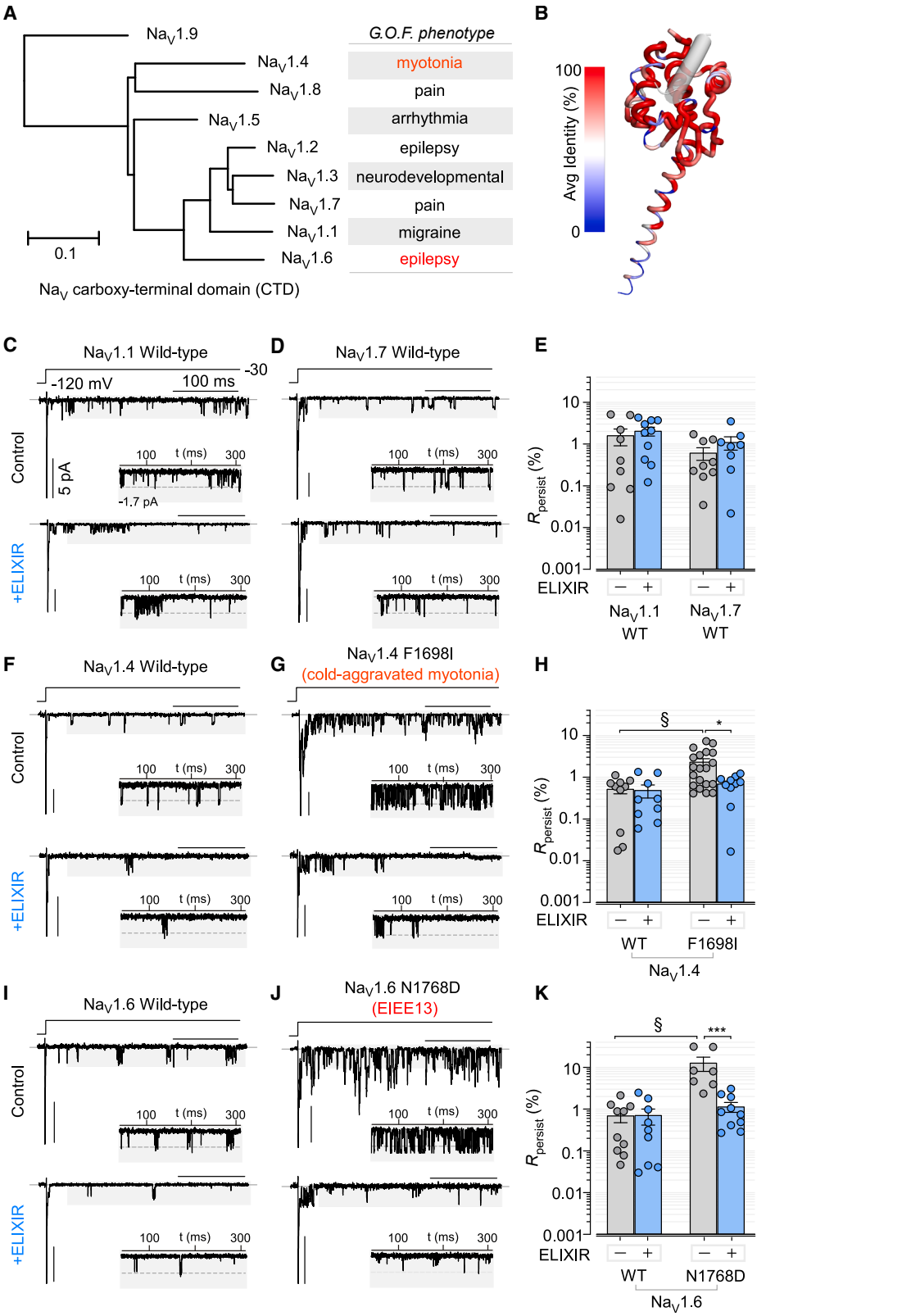
(E) Bar graph summarizes effect of ELIXIR on $R_{persist}$ for various mutations or with phosphorylation. ΔKPQ without (n = 17 cells, 995 sweeps) and with ELIXIR (n = 10 cells, 936 sweeps), F1759A without (n = 17 cells, 1,083 sweeps) and with ELIXIR (n = 7 cells, 725 sweeps), IQ/AA without (n = 15 cells, 926 sweeps) and with ELIXIR (n = 10 cells, 977 sweeps), WT $Na_V1.5$ + PKA_{Cat} without (n = 10 cells, 710 sweeps) and with ELIXIR (n = 10 cells, 841 sweeps), and $Na_V1.5$ + $CaMKII_{T286D}$ without (n = 11 cells, 1,033 sweeps) and with +ELIXIR (n = 10 cells, 1,029 sweeps). Each bar, mean ± SEM. ****p < 0.0001 (ΔKPQ, F1759A), **p = 0.0015 (IQ/AA), *p = 0.0101 (+PKA_{Cat}), *p = 0.0101 ($CaMKII_{T286D}$) by a two-tailed Mann-Whitney U test.

(F) Schematic shows ELIXIR conjugated to a cell-penetrating peptide (ELIXIR-cpp). I_{NaL} of $Na_V1.5$ ΔKPQ following treatment with different concentrations of ELIXIR-cpp. The IC_{50} and n (Hill coefficient) values were obtained through a global fit of the data with a Hill equation. Each dot, mean ± 95% CI. O.E., I_{NaL} with ELIXIR overexpression.

See also Figure S5.

Accordingly, we sought to determine the effect of ELIXIR on I_{NaL} of selected wild-type Na_V channels expressed in distinct tissues: (1) $Na_V1.1$ in neurons, (2) $Na_V1.4$ in skeletal muscle, (3) $Na_V1.6$ in neurons, and (4) $Na_V1.7$ in peripheral nervous system neurons. When expressed in HEK293 cells, these channels exhibited varying amplitudes of I_{NaL}, with a range of $R_{persist}$ values. Co-expression of ELIXIR failed to evoke a statistically significant change in the basal I_{NaL} of any of the wild-type Na_V isoforms examined: $Na_V1.1$ (Figures 4C, 4E, and S6I), $Na_V1.7$ (Figures 4D, 4E, and S6J), $Na_V1.4$ (Figures 4F, 4H, and S6K), or $Na_V1.6$ (Figures 4I, 4K, and S6M).

Thus affirmed, we probed whether ELIXIR could modulate I_{NaL} of mutant channels that have deficits in inactivation (i.e., if the IG has reduced propensity for occupying the pore-proximal site). For $Na_V1.4$, increased I_{NaL} is considered a relevant pathophysiological mechanism for congenital myotonia.[6] We considered the effect of ELIXIR on an $Na_V1.4$ variant (F1698I) associated with cold-aggravated myotonia in humans.[57,58] When expressed alone, $Na_V1.4$ F1698I had a small (~4-fold) but statistically significant increase in $R_{persist}$ compared with wild-type $Na_V1.4$ (Figures 4F–4H, S6K, and S6L). Co-expression of ELIXIR reversed I_{NaL} of the mutant channel to near wild-type levels (Figures 4F–4H and S6L), despite the wild-type channel being unperturbed by ELIXIR. Gene variants in voltage-gated sodium channels are a leading cause of monogenic epilepsies.[59] Variants in *SCN8A*, encoding $Na_V1.6$, increase neuronal excitability by impairing inactivation and increasing I_{NaL}.[21,60,61] Consistent with previous findings,[21,62–64] analysis of an $Na_V1.6$ variant (N1768D) linked to epilepsy and seizure disorders showed a markedly elevated I_{NaL} as compared with the wild-type channel (Figures 4I–4K, S6M, and S6N). Strikingly, co-expression of ELIXIR reduced I_{NaL} (~10-fold) of $Na_V1.6$ N1768D to near wild-type levels (Figures 4I–4K, S6M, and S6N). These findings suggest that although ELIXIR may bind multiple Na_V isoforms, its inhibition of I_{NaL} is most prominent when there are defects in the inactivation process. This feature may allow ELIXIR to preferentially modulate mutant or pathologically modified channels.



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ELIXIR reverses altered neuronal excitability linked to $Na_V1.6$ channelopathy

As ELIXIR potently inhibited the I_{NaL} of the $Na_V1.6$ N1768D variant, we considered whether it can rectify the alteration of neuronal AP properties associated with this variant.[64] To do so, we tested the effect of ELIXIR in a mouse model of neurodevelopmental disorder expressing the $Na_V1.6$ N1768D variant.[65] Previous studies have shown that these mice develop seizures and present with SUDEP,[65] consistent with clinical phenotypes.[21] These mice also display aberrant neuronal AP spike properties, including AP prolongation, spike frequency adaptation, and depolarization block, reflecting an increased I_{NaL} in various neuronal subtypes.[63,64,66,67] We packaged ELIXIR with an mCherry expression reporter into an adeno-associated virus 9 (AAV9) vector and used intracerebroventricular injection for delivery to cortical neurons of P0 N1768D homozygous knockin (*SCN8A*$^{N1768D/N1768D}$) and wild-type littermate control (*SCN8A*$^{+/+}$) mice. We targeted layer V pyramidal neurons for whole-cell patch-clamp electrophysiology recordings in brain slices. Fluorescence imaging confirmed expression of ELIXIR (Figure 5A). Comparison of wild-type and N1768D neurons confirmed a prolongation of the APD (Figures 5B, 5C, S7A, and S7B), due to a slowing of the repolarizing phase (Figures S7C and S7D).[64] ELIXIR-positive neurons in N1768D slices exhibited a reduced APD compared with ELIXIR-negative N1768D neurons (Figures 5B, 5C, and S7B). The APs of wild-type neurons were found to be largely insensitive to ELIXIR (Figures 5B, 5C, and S7B). Analysis of AP properties showed a small change in AP firing rate in neurons from wild-type mice upon ELIXIR expression (Figures 5D, 5F, and S7E). By contrast, neurons from N1768D mice displayed premature depolarization block, which may contribute to disease phenotypes in *SCN8A* neurodevelopmental disorders.[67] ELIXIR expression significantly attenuated depolarization block (Figures 5E, 5G, S7E, and S7F). These results reveal that ELIXIR is effective at reversing the altered neuronal AP properties in $Na_V1.6$ GoF channelopathies linked to epilepsy.

ELIXIR reduces I_{NaL} and APD_{90} of CaMKII$\delta_{C,OE}$ ventricular myocytes

In HF, phosphorylation of $Na_V1.5$ by CaMKII results in elevated I_{NaL}, which contributes to increased risk of cardiac arrhythmias.[51,52,68] We examined whether ELIXIR may reverse pathological I_{NaL} in adult mouse ventricular myocytes (aMVMs) isolated from a transgenic mouse model that overexpresses the predominant cardiac CaMKII isoform (CaMKII$\delta_{C,OE}$).[69] Previous studies have shown that overexpression of CaMKIIδ_C induced cardiac hypertrophy and dilated cardiomyopathy, leading to arrhythmogenic HF and premature death in these mice.[69,70] Indeed, we observed morphological and electrophysiological changes in the hearts and cardiomyocytes of the CaMKII$\delta_{C,OE}$ mice (Figures S8A–S8F) that were indicative of hypertrophy and consistent with HF. We utilized ELIXR-cpp for short-term intracellular delivery of ELIXIR into aMVMs (ELIXIR-cpp; Figure 6A). Flow cytometric analysis of non-transgenic (nTG) aMVMs treated with ELIXIR-cpp showed a concentration-dependent increase in fluorescein isothiocyanate (FITC) fluorescence intensity (Figures 6B and 6C), confirming uptake of ELIXIR-cpp by the myocytes. I_{NaL} was quantified through whole-cell patch-clamp recordings as the magnitude of Na^+ current following 500 ms of depolarization normalized to the whole-cell capacitance. In line with previous studies,[68] the CaMKII$\delta_{C,OE}$ aMVMs displayed a 2-fold increase in I_{NaL} relative to aMVMs from nTG animals (Figures 6D–6F). Application of ELIXIR-cpp caused a small but statistically significant reduction in the I_{NaL} of nTG myocytes (Figures 6D and 6F). By comparison, incubation with ELIXIR-cpp had a much stronger effect on the I_{NaL} of CaMKII$\delta_{C,OE}$ aMVMs, which was reduced to near wild-type levels (Figures 6E and 6F).

To probe the functional consequences of ELIXIR-cpp on cardiomyocyte function, we undertook current clamp recordings to quantify changes in AP morphology. Consistent with the increase in I_{NaL}, the APD of CaMKII$\delta_{C,OE}$ myocytes was prolonged relative to that of nTG myocytes (Figures 6G–6I). Moreover, myocytes from the CaMKII$\delta_{C,OE}$ mice had greater variability in the APD, quantified as short-term variability (STV[71,72]; Figures 6J–6L), than those from nTG animals. Application of ELIXIR-cpp yielded a significant reduction in the APD (Figures 6H and 6I) and STV (Figures 6K and 6L) of aMVMs from CaMKIId$_{C,OE}$ mice, reversing both to near nTG levels. By comparison, ELIXIR-cpp caused no change in APD or STV of nTG aMVMs. Collectively, these findings illustrate the ability of ELIXIR, applied extracellularly as a peptide, to tune native channel function.

ELIXIR inhibits I_{NaL} in LQT3 patient-derived iPSC-CMs

While ELIXIR-cpp allows for short-term modification of Na_V function in cardiomyocytes, constitutive expression of ELIXIR may be advantageous in pathophysiological settings. As such, we used

Figure 4. Selectivity of ELIXIR for pathogenic late Na^+ current

(A) Phylogenetic tree of the nine human Na_V CTD sequences with GoF phenotypes.

(B) ColabFold model of ELIXIR (gray) with $Na_V1.5$ CTD with thickness and color corresponding to the average sequence identity across Na_V channels.

(C and D) Exemplar recordings of WT $Na_V1.1$ (C) and $Na_V1.7$ (D) without (top) and with (bottom) ELIXIR.

(E) Bar graph shows effect of ELIXIR on $Na_V1.1$ and $Na_V1.7$. Each bar, mean ± SEM, $Na_V1.1$ without (n = 9 cells; 912 sweeps) and with ELIXIR (n = 10 cells; 880 sweeps) and $Na_V1.7$ without (n = 9 cells; 833 sweeps) and with ELIXIR (n = 8 cells; 765 sweeps).

(F and G) Exemplar recordings show the effect of ELIXIR on WT $Na_V1.4$ (F) and the F1698I mutant (G).

(H) Bar graph summarizes effect of ELIXIR on WT and mutant $Na_V1.4$ channels. Each bar, mean ± SEM. $Na_V1.4$ without (n = 10 cells, 953 sweeps) and with ELIXIR (n = 9 cells, 812 sweeps) and F1698I mutant (n = 20 cells, 1,647 sweeps) and with ELIXIR (n = 10 cells, 956 sweeps). §p = 0.0101 (WT versus F1698I), *p = 0.0118 (F1698I with and without ELIXIR) by a one-way ANOVA followed by a Tukey's multiple comparisons test.

(I and J) Exemplar recordings show the effect of ELIXIR on WT $Na_V1.6$ (I) and N1768D mutant (J). EIEE13, early infantile epileptic encephalopathy 13.

(K) Bar graph summarizes effect of ELIXIR on WT and mutant $Na_V1.6$. Each bar, mean ± SEM. $Na_V1.6$ without (n = 10 cells, 973 sweeps) and with ELIXIR (n = 9 cells, 885 sweeps) and $Na_V1.6$ N1768D mutant without (n = 7 cells, 876 sweeps) and with ELIXIR (n = 10 cells, 1,162 sweeps). §p = 0.0009 (WT versus N1768D), *p = 0.0008 ($Na_V1.6$ N1768D mutant with and without ELIXIR) by one-way ANOVA followed by a Tukey's multiple comparisons test.

See also Figure S6.

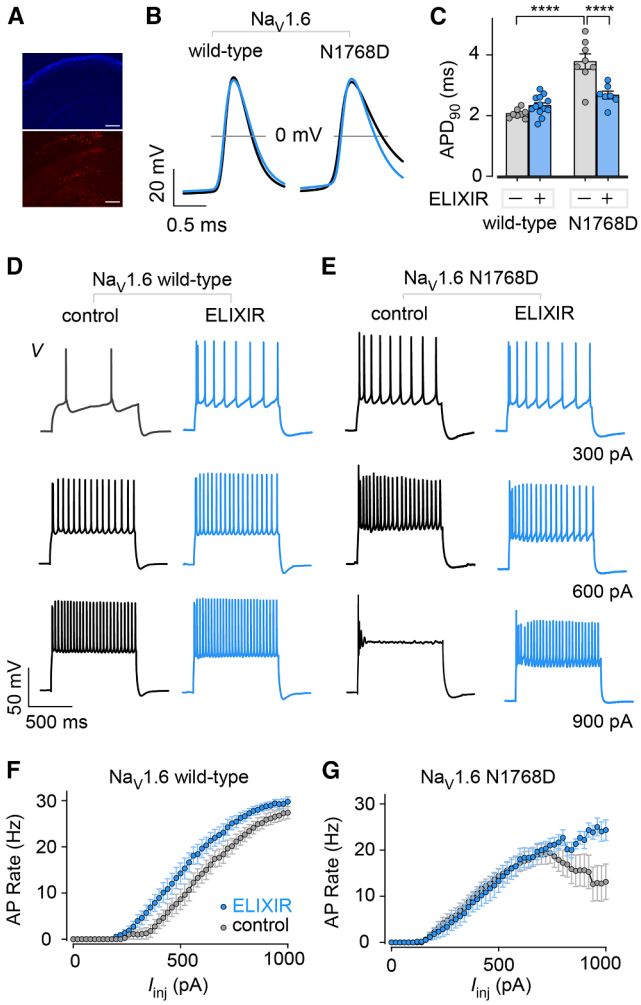


Figure 5. ELIXIR rescues Na$_V$ channelopathy-induced AP abnormalities in mouse brain

(A) AAV9-mediated expression of ELIXIR in neurons of the cerebral cortex. Top, DAPI; bottom, mCherry, expression marker. Scale bar, 500 μm.

(B) Group average of AP recordings from control (black trace) and ELIXIR-expressing (blue) neurons from WT Na$_V$1.6 (left) and Na$_V$1.6 N1768D (right) mice.

(C) Bar graph summarizing the effect of ELIXIR on the APD$_{90}$ of control and ELIXIR-expressing WT Na$_V$1.6 (*SCN8A*$^{+/+}$; control: n = 8 cells from 2 mice; ELIXIR: n = 8 cells from 2 mice) or Na$_V$1.6 N1768D (*SCN8A*$^{N1768D/N1768D}$, control: n = 8 cells from 2 mice; ELIXIR: n = 7 cells from 2 mice) neurons. ****$p < 0.0001$ by two-way ANOVA followed by Tukey's multiple comparisons test.

(D and E) Exemplar AP recordings from WT (D) or N1768D (E) neurons with or without ELIXIR.

(F and G) Input-output curves depict the AP rate of WT neurons (F; control: n = 8 cells from 2 mice; ELIXIR: n = 8 cells from 2 mice) and N1768D neurons (G; control: n = 8 cells from 2 mice; ELIXIR: n = 6 cells from 2 mice) in the presence or absence of ELIXIR as a function of applied current.

See also Figure S7.

AAV9 to express ELIXIR in two cardiac disease models: (1) human cardiomyocytes differentiated from induced pluripotent stem cells (hiPSC-CMs) and (2) transgenic mice expressing a mutant Na$_V$1.5 (F1759A^{tg}; Figure 7A).

First, we considered hiPSC-CMs derived from an LQT3 patient heterozygous for the Na$_V$1.5 ΔKPQ variant[26] and from a healthy donor (HD).[73] Following differentiation and maturation, spontaneous contractions were observed in both the HD and ΔKPQ hiPSC-CMs. Multichannel recordings of the HD hiPSC-CMs revealed low baseline levels of I_{NaL} that were minimally altered by AAV9-expression of ELIXIR (Figures 7B and S8G). By contrast, the ΔKPQ hiPSC-CMs exhibited a markedly increased I_{NaL} (Figures 7C, 7D, and S8H), which was reduced upon expression of ELIXIR (Figures 7C, 7D, and S8H). This underscores the efficacy of ELIXIR in diminishing I_{NaL} linked to LQT3 variants in the more pathophysiologically relevant setting of human cells.

Second, we considered transgenic mice that inducibly express the lidocaine-resistant Na$_V$1.5 F1759A mutant channel. These mice have been previously reported to have an elevated I_{NaL} and prolonged QT interval.[2] Subcutaneous injections delivered an AAV9 encoding ELIXIR or GFP to P4 mice. 8 weeks after the injection of the virus, ventricular cardiomyocytes were isolated, and ELIXIR expression was confirmed by confocal imaging of mCherry fluorescence (Figure 7E). Multichannel recordings of cardiomyocytes from GFP-injected mice revealed high basal I_{NaL} consistent with previous reports[25] (Figure 7F). By comparison, cardiomyocytes from ELIXIR-injected animals showed a significantly attenuated I_{NaL} (Figures 7F and 7G). Increased I_{NaL} from Na$_V$1.5 manifests as a prolonged QT interval on the electrocardiogram (ECG) and is the basis for LQT3 in humans. As such, we obtained limb-lead surface ECG recordings from both ELIXIR-injected mice and GFP-control littermates (Figure 7H). QT intervals, corrected for heart rate using Mitchell's formula (QT$_C$),[74] were significantly shorter in the ELIXIR-expressing mice than in the GFP-expressing control animals (Figure 7I). We observed no significant differences in the RR or QRS intervals (Figure 7I). These results confirm the effectiveness of long-term ELIXIR expression in tuning endogenous I_{NaL} in both human and murine models of Na$_V$1.5 dysfunction.

DISCUSSION

Strategies to precisely manipulate ion channel function are highly sought after both as tools for unraveling their physiological functions and for developing therapies.[75] A prominent example involves Na$_V$ channel family (Na$_V$1.1–1.9), where GoF variants that diminish inactivation are associated with cardiac arrhythmia,[76] myotonia,[6] and epilepsy.[21] By combining structural insights and a computational framework, we developed a peptide modulator called ELIXIR that promotes Na$_V$ inactivation. ELIXIR binds to its desired interface (Na$_V$1.5$_{CTD}$) with submicromolar affinity and modulates pathological I_{NaL} in multiple disease models. These findings illustrate the utility of *de novo* protein design for engineering ion channel modulators in health and disease.

Many ion channels are large membrane-spanning multi-subunit complexes whose gating depends on intricate conformational dynamics. To date, the prevailing strategy to engineer ion channel modulators has been empirical, focused on repurposing known ion channel modulators[26,77] or developing nanobodies that interact with specific domains.[78] Computational protein design furnishes a powerful avenue to "write" modulatory proteins with desired functions. Advances in this methodology,

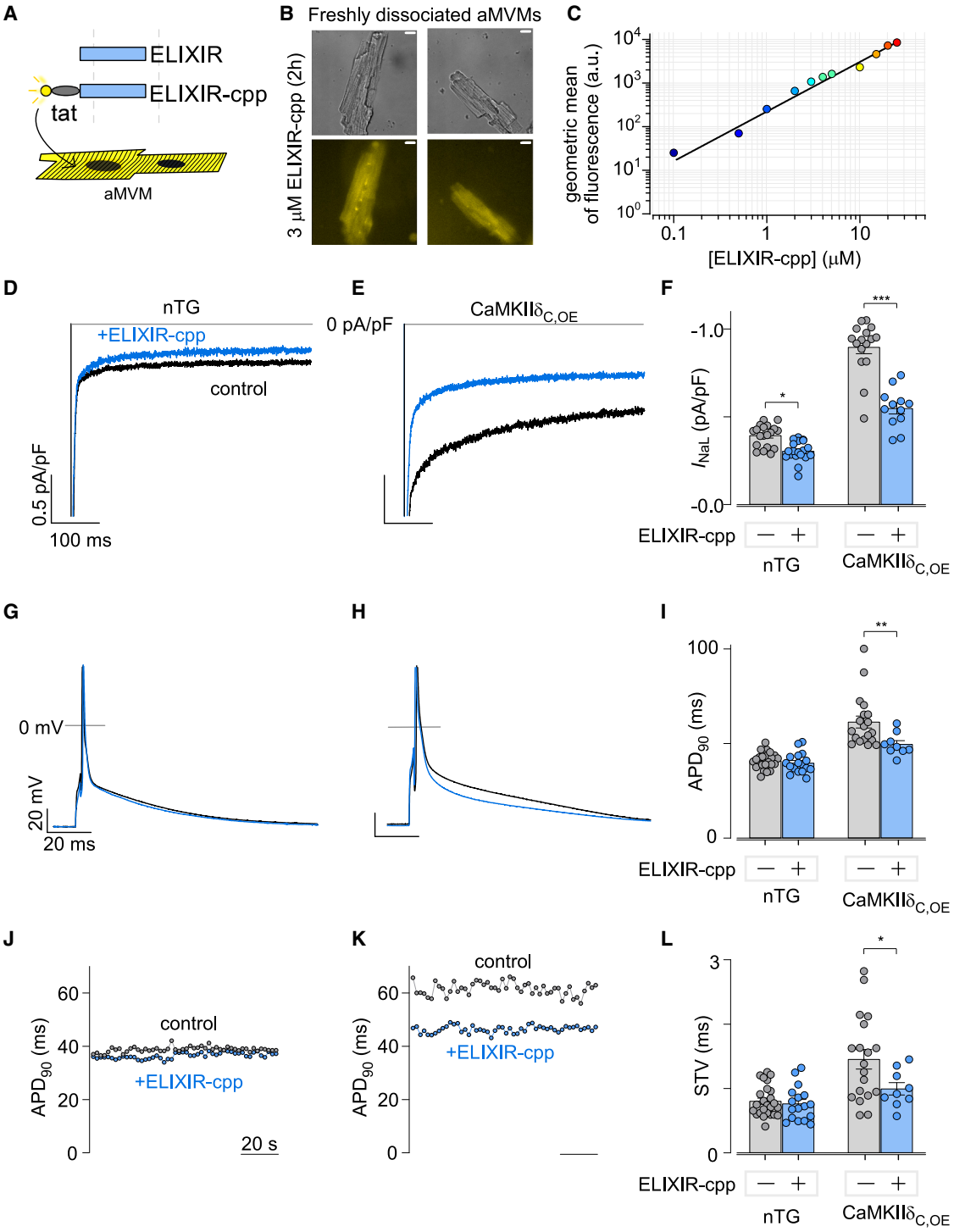


Figure 6. ELIXIR inhibits I_{NaL} and reduces APD_{90} in murine ventricular cardiomyocytes

(A) Schematic shows design of cell-penetrating ELIXIR (ELIXIR-cpp).

(B) Epifluorescence images show uptake of ELIXIR-cpp into isolated ventricular myocytes from adult mice. Scale bar, 10 μm.

(C) Flow cytometric analysis shows uptake of ELIXIR-cpp into aMVMs.

(D) Exemplar whole-cell recordings show I_{NaL} in freshly dissociated aMVM from nTG mice at baseline (black) or with 3 μM ELIXIR-cpp (blue).

(E) Exemplar whole-cell recordings show I_{NaL} inhibition in aMVM from CaMKIIδ$_{C,OE}$ transgenic mice that develop HF. Format as in (D).

(legend continued on next page)

particularly deep-learning approaches, have improved accuracy and reduced the required computational time.[27–29,43,79] The availability of open-source and easy-to-use platforms has increased the accessibility of these methods to the broader scientific community. While promising, experimental validation of *de novo* sequences has been limited. For *in silico* designed *de novo* protein binders, a small number of targets have been tested *in vitro*[28,29] and even fewer *in situ* or *in vivo*.[80] Our results demonstrate the effectiveness of these methods for developing ion channel modulators that function *in vivo* and further highlight the general utility of currently available algorithms.

In the heart, multiple pathophysiological mechanisms lead to increased I_{NaL}. These include (1) human $Na_V1.5$ variants,[81] (2) phosphorylation by CaMKII or PKA,[51,52] (3) ROS[52] or cellular metabolites,[82] (4) altered binding of interacting proteins,[48,83–85] and (5) diverse signaling mechanisms.[86–88] Considerable effort has been devoted to developing selective I_{NaL} inhibitors. Class 1b antiarrhythmics (e.g., lidocaine and mexiletine), which bind with rapid kinetics to a transmembrane site in $Na_V1.5$, exhibit preference for inhibiting I_{NaL}.[9] Structurally similar to lidocaine, ranolazine exhibits a greater selectivity for I_{NaL} inhibition. High-resolution structures[23,24] show that these molecules bind to a conserved site in the channel pore. This shared binding site, however, limits the selectivity of these molecules for late versus peak Na^+ current, as their interaction obstructs ion flux through the pore. Instead, ELIXIR relies on a natural inactivation mechanism by enhancing the release of the IG from the $Na_V1.5$ CTD through a competitive scheme. This strategy has two advantages. First, by not targeting the channel pore or the VSDs, ELIXIR does not significantly perturb channel activation. Second, ELIXIR is unlikely to modify channels with no baseline defect in inactivation. Specifically, for wild-type channels, the domain IV (DIV) VSD movement is considered a rate-limiting step for both the development of and recovery from inactivation.[34] Without the movement of DIV VSD, it is unlikely that the pore-proximal site is available for interaction with the IG. This limits the action of ELIXIR on wild-type channels. One interesting possibility, hinted at by our results, is that in pathophysiological settings, the release of IG from CTD may become a limiting step such that promoting its release would boost inactivation. This aspect may lend ELIXIR selectivity for tuning pathogenic I_{NaL}, a desirable feature from a translational perspective. A corollary is disease-causing perturbations in Na_V channels converge on a common endpoint—the diminished ability of the IG to bind the pore-proximal site. For some $Na_V1.5$ perturbations examined, ELIXIR did not fully reverse the increase in I_{NaL}. This partial inhibition may reflect incomplete disruption of the IG-CTD interaction by ELIXIR or other defects caused by these perturbations. Further optimization of ELIXIR through affinity maturation or the production of mutation-specific binders could address this. We previously found that the Na_V-interacting protein FHFs also inhibit I_{NaL}.[26] We engineered a 39-aa $FHF1_A$ fragment (FHF-inhibiting x Region, FixR) as an I_{NaL} inhibitor. While the molecular mechanism of FHFs/FixR remains unclear, it differs from that of ELIXIR, as FHFs promote inactivation of mutant $Na_V1.5$ Δ1810 that lacks the ELIXIR-binding pocket.[26] It is possible that the two peptides may be beneficial in different settings. As FHFs are natively present in various tissues, the effect of FixR may vary depending on the presence of endogenous FHFs. By contrast, ELIXIR as a *de novo* peptide may be broadly applicable.

Na_V malfunction has been implicated in various diseases, including epilepsy,[64] myotonia,[6] and chronic pain.[22] The high degree of sequence conservation in the Na_V EFL suggests that ELIXIR may be effective for GoF channelopathies across the Na_V family, an attractive possibility given the range of diseases associated with dysfunction of these channels. Indeed, we found that ELIXIR sufficed to reverse the increase in I_{NaL} observed with channelopathic mutations in skeletal muscle $Na_V1.4$ and neuronal $Na_V1.6$. Moreover, expression of ELIXIR in mouse brain sufficed to partially reverse changes in neuronal AP morphology and excitability. More broadly, the Na_V CTD has an architecture similar to the C-termini of other ion channels, including the voltage-gated calcium channels[15] and the sodium leak channel (NALCN).[14] In high-resolution structures of these channels, the EFL region of the CTD is bound to the domain III-IV linker,[14,15] hinting at conserved modulatory mechanisms. If so, our design strategy may be helpful for developing selective actuators of this physiologically important ion channel superfamily.

Although ion channel dysfunction has long been associated with life-threatening diseases, these proteins have been challenging targets for drug development.[89] Our results demonstrate the effectiveness of computational protein design for the development of synthetic proteins/peptides that tune channel dynamics. This ability could turbocharge the creation of innovative strategies to directly target the pathophysiology of multiple channelopathies and enable precision engineering of nature's transistors.

Limitations of the study

A high-resolution structure of ELIXIR bound to the $Na_V1.5$ channel is not available. Nevertheless, structural predictions of ELIXIR/CTD complexes by various algorithms all converge to a similar interface, which is consistent with mutational analysis.

(F) Bar graph summarizes the effect of ELIXIR on I_{NaL} in aMVM from nTG versus $CaMKII\delta_{C,OE}$ mice. Each bar, mean ± SEM. nTG without (n = 19 cells from 6 mice) or with ELIXIR-cpp (n = 19 cells from 6 mice) and $CaMKII\delta_{C,OE}$ without (n = 16 cells from 5 mice) or with ELIXIR-cpp (n = 12 cells from 5 mice). *p = 0.034, ***p < 0.001 by two-way ANOVA followed by a Tukey's multiple comparisons test.

(G and H) Exemplar AP recordings from nTG (G) and $CaMKII\delta_{C,OE}$ (H) aMVMs both without (black) and with (blue) ELIXIR.

(I) Bar graph summarizes the ELIXIR effect on APD_{90}. Each bar, mean ± SEM. nTG without (n = 25 cells from 6 mice) or with ELIXIR-cpp (n = 17 cells from 6 mice) and $CaMKII\delta_{C,OE}$ without (n = 19 cells from 5 mice) or with ELIXIR-cpp (n = 9 cells from 4 mice) **p = 0.004 by two-way ANOVA followed by Tukey's multiple comparisons test.

(J and K) Diary plot shows variability in APD_{90} for nTG or $CaMKII\delta_{C,OE}$ transgenic mice with and without ELIXIR.

(L) Bar graph summarizes changes in short-term variability (STV) of APDs. Each bar, mean ± SEM. nTG without (n = 25 cells from 6 mice) or with ELIXIR-cpp (n = 17 cells from 6 mice) and $CaMKII\delta_{C,OE}$ without (n = 19 cells from 5 mice) or with ELIXIR-cpp (n = 9 cells from 4 mice).

See also Figure S8.

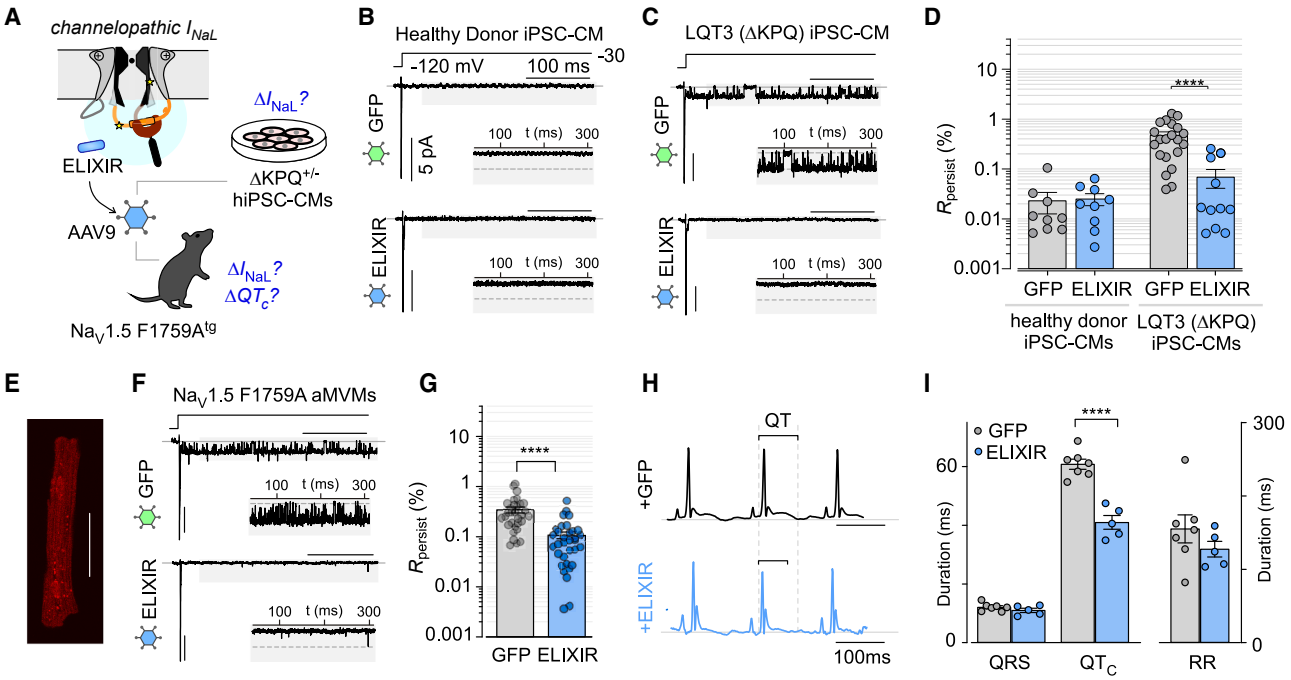


Figure 7. ELIXIR reverses $Na_V1.5$ dysfunction in LQT3 hiPSC-CMs and in F1759A^{tg} mice

(A) Schematic shows AAV9-expression of ELIXIR into cultured LQT3 hiPSC-CMs and F1759A^{tg} mice to determine changes in I_{NaL} or ECG.

(B and C) Exemplar recordings of iPSC-CMs from a healthy donor (HD, B) or an LQT3 patient (ΔKPQ variant) (C) transduced with GFP (top) or ELIXIR (bottom) AAV9.

(D) Bar graph summarizes the effect of ELIXIR on I_{NaL} in iPSC-CMs. Each bar, mean ± SEM. HD iPSC with GFP (n = 9 cells; 781 sweeps from 2 differentiations) or ELIXIR (n = 9 cells; 803 sweeps from 2 differentiations) and LQT3 (ΔKPQ) iPSC-CMs with GFP (n = 21 cells; 1,361 sweeps from 2 differentiations) or ELIXIR (n = 11 cells; 885 sweeps from 2 differentiations). ****$p < 0.0001$ by a two-tailed Mann-Whitney U test.

(E) Confocal image shows mCherry expression as a marker of ELIXIR production by ventricular myocytes freshly isolated from adult $Na_V1.5$ F1759A mice. Scale bar, 50 μm.

(F) Exemplar multichannel recordings from ventricular myocytes isolated from $Na_V1.5$ F1759A mice infected with either GFP (top) or ELIXIR (bottom).

(G) Bar graph summarizes effect of ELIXIR on I_{NaL} of $Na_V1.5$ F1759A aMVMs. Each bar, mean ± SEM. $Na_V1.5$ F1759A mice injected with GFP AAV (n = 31 cells, 2,181 sweeps from 6 mice) or ELIXIR AAV (n = 33 cells, 2,396 sweeps from 6 mice). ****$p < 0.0001$ by a two-tailed Mann-Whitney U test.

(H) Exemplar limb-lead surface ECGs of isoflurane-anesthetized F1759A mice with AAV9-expression of GFP (top) or ELIXIR (bottom).

(I) Bar graph comparison of the RR (H) and QRS and QT_C (I) intervals of isoflurane-anesthetized F1759A mice infected with an AAV containing GFP (n = 7 mice) or ELIXIR (n = 5 mice). ****$p < 0.0001$ by nested t test.

See also Figure S8.

With viral expression, we were unable to estimate the concentration of ELIXIR as the mCherry marker is bicistronically expressed. It is possible that the concentration of ELIXIR may be supramaximal, which may cause off-target effects. Future studies will consider the dose-dependence of ELIXIR expression. More broadly, although peptides are used clinically (e.g., semaglutide and insulin), the translational potential of ELIXIR is limited by difficulty in intracellular delivery. Our studies utilized a cell-penetrating peptide and viral delivery of ELIXIR. Nevertheless, these strategies have limitations, including immunogenicity. Future studies will explore other delivery approaches. It may also be possible to identify small molecules that interact with $Na_V1.x$ CTD and mimic the action of ELIXIR.

RESOURCE AVAILABILITY

Lead contact

Requests for resources, reagents, and further information should be directed to and will be fulfilled by the lead contact, Manu Ben-Johny (manu.ben.johny@columbia.edu).

Materials availability

All unique reagents generated in this study are available upon request from the lead contact.

Data and code availability

The peptide design protocol is available at https://colab.research.google.com/github/sokrypton/ColabDesign/blob/v1.1.1/af/examples/peptide_binder_design.ipynb.

All data are included in the main text or supplemental information. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

R.M. and M.B.-J. conceptualized and designed the project. S.O. developed the protocol for peptide hallucination. R.M., B.H., E.R.C., T.M.C., A.R.R., L. F., M.Y., L.Y., B.-X.C., A.N.K., N.C., R.J., and M.B.-J. performed experiments. R.M., B.H., E.C., L.F., N.C., and M.B.-J. analyzed data. R.M., E.C., B.H., R.J., E.Y.W., S.O.M., J.K., C.D.M., D.M.B., and M.B.-J. contributed reagents/analytic tools. S.O.M., S.O., J.K., C.D.M., D.M.B., and M.B.-J. acquired funding. R.M., B.H., and M.B.-J. made figures and wrote the original manuscript draft. All authors revised the manuscript.

DECLARATION OF INTERESTS

R.M. and M.B.-J. are inventors on a provisional patent filed by Columbia University (attorney docket no. 44010.196US-PROV//CU24196) on the application of ELIXIR.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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  - $CaMKII\delta_{C,OE}$ mice
  - $Na_V1.5$ $F1759A^{tg}$ mice
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- METHOD DETAILS
  - Molecular biology
  - Peptide design
  - Protein expression and purification
  - Fluorescence anisotropy
  - Whole-Cell recordings from HEK293 cells
  - Cell-attached Multichannel recordings
  - Flow cytometric FRET 2-hybrid analysis
  - Phylogenetic tree construction
  - Adeno-associated virus generation
  - Flow cytometric analysis of ELIXIR uptake
  - Brain slice electrophysiology
  - Enzymatic isolation of ventricular $CaMKII\delta_{C,OE}$ cardiomyocytes
  - $CaMKII\delta_{C,OE}$ cardiomyocyte electrophysiology
  - Confocal imagining
  - ECG analysis
  - Images of protein structures
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
Escherichia coli BL21 (DE3+)	Agilent	Cat#230280
Escherichia coli XL-10 Gold	Agilent	Cat#200315
pAAV[Exp]-CAG>EGFP:WPRE (GFP AAV9)	VectorBuilder	Cat# VB010000-9287ffw
pAAV[Exp]-CMV>{ELIXIR 1}:WPRE (GFP AAV9)	This Paper	N/A
Chemicals, peptides, and recombinant proteins		
ELIXIR-cpp	This Paper	N/A
Ranolazine	Sigma Aldrich	Cat#R6152
Cycloheximide	Tocris	Cat#0970/100
Experimental models: Cell lines		
Human: HEK293 Cells	ATCC	Cat#CRL-1573; RRID:CVCL_0045
Human: hiPSC-Healthy Donor Cell line	Huang et al.[73]	N/A
Human: hiPSC-LQT3 Cell line	Chakouri et al.[26]	N/A
Experimental models: Organisms/strains		
Mouse: C57BL/6J SCN8A-N1768D	Wagnon et al.[65]	N/A
Mouse: C57BL/6J CaMKII$\delta_{C,OE}$	Zhang et al.[69]	N/A
Mouse: B6CBA/F2/ FVB/N SCN5A-F1759A (αMHC-rtTA x tetO-Na$_V$1.5 F1759A)	Wan et al.[2]	N/A
Recombinant DNA		
pGW-WT Na$_V$1.5	Chakouri et al.[26]	N/A
pGW-Na$_V$1.5 D1864	This Paper	N/A
pGW-Na$_V$1.5 D1810	Chakouri et al.[26]	N/A
pGW-Na$_V$1.5 DKPQ	Chakouri et al.[26]	N/A
pGW-Na$_V$1.5 IQ/AA	Chakouri et al.[26]	N/A
TetO-Na$_V$1.5 F1759A	Wan et al.[2]	N/A
pGEX-Na$_V$1.5$_{CTD}$	This Paper	N/A
pET24b-CaM	Gabelli et al.[45]	N/A
pIRES2-ELIXIR$_{Original}$	This Paper	N/A
pIRES2-ELIXIR	This Paper	N/A
pIRES2-ELIXIR L8A	This Paper	N/A
pIRES2-ELIXIR F12E	This Paper	N/A
pIRES2-ELIXIR Y16A	This Paper	N/A
pCDNA3-Ven-Na$_V$1.5$_{CTD}$	This Paper	N/A
pCDNA3-Cer-ELIXIR	This Paper	N/A
pCDNA3-Cer-ELIXIR L8A	This Paper	N/A
pCDNA3-Cer-ELIXIR F12E	This Paper	N/A
pCDNA3-Cer-ELIXIR-Y16A	This Paper	N/A
pEGFP C1-Venus	Koushik et al.[90]	Addgene: Cat#27794; RRID:Addgene_27794
pEGFP C1-Cerulean	Koushik et al.[90]	Addgene: Cat#27796; RRID:Addgene_27796
pEGFP C1-Cerulean-5AA-Venus	Koushik et al.[90]	Addgene: Cat#26394; RRID:Addgene_26394
pEGFP C1-Cerulean-32AA-Venus	Koushik et al.[90]	Addgene: Cat#26396; RRID:Addgene_26396
pEGFP C1-Cerulean-Traf-Venus	Koushik et al.[91]	Addgene: Cat#27803; RRID:Addgene_27803
pCDNA3.1-PKA$_{Cat}$	Lee et al.[92]	N/A
pCDNA3.1-CaMKII$_{T286D}$	Chakouri et al.[26]	N/A
pIR-CMV-IRES-mScarlet-WT Na$_V$1.1	DeKeyser et al.[93]	N/A

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
pCDNA3.1-WT $Na_V1.4$	Ben-Johny et al.[94]	N/A
pCDNA3.1-$Na_V1.4$ F1698I	Ben-Johny et al.[94]	N/A
pCDNA4TO-IRES-mScarlet-WT $Na_V1.6$	DeKeyser et al.[93]	Addgene: Cat#162280; RRID: Addgene_162280
pCDNA4TO-IRES-mScarlet-$Na_V1.6$ N1768D	This Paper	N/A
pIR-CMV-SCN1A-Variant-1-IRES-mScarlet	DeKeyser et al.[93]	Addgene: Cat#162278; RRID: Addgene_162278
pCMV-WT $Na_V1.7$	This Paper	N/A
Software and algorithms		
AfDesign	This Paper	https://colab.research.google.com/github/sokrypton/ColabDesign/blob/v1.1.1/af/examples/peptide_binder_design.ipynb
MATLAB	MathWorks	https://www.mathworks.com/products/matlab.html; RRID:SCR_001622
SCLate (Late current analysis)	Abrams et al.[25]	https://github.com/manubenjohny/LateCurrent
FACS_FRET (Flow Cytometric FRET analysis)	Rivas et al.[95]	https://github.com/manubenjohny/FACS_FRET
PyMOL	Schrödinger LLC	https://pymol.org/; RRID:SCR_000305
ColabFold	Mirdita et al.[41]	https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb
ESMFold	Lin et al.[42]	https://esmatlas.com/resources?action=fold
OmegaFold	Wu et al.[27]	https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/beta/omegafold.ipynb
RoseTTAFold	Baek et al.[43]	https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/RoseTTAFold.ipynb
COCOMAPS	Vangone et al.[96]	http://www.molnac.unisa.it/BioTools/cocomaps
GraphPad Prism	GraphPad	https://www.graphpad.com/; RRID:SCR_002798

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Mammalian cell lines

HEK293 cells (ATCC CRL1573) were cultured in 60-mm dishes on glass coverslips and transfected with the Ca^{2+}-phosphate method. For electrophysiology recordings, cells were co-transfected with 4-10 μg of cDNA encoding the desired Na_V channel variant, 1 μg of YFP, and 0.5 μg of simian virus 40 T-antigen ± 3 μg of ELIXIR. For experiments investigating phosphorylation-dependent late Na^+ current, 3 μg of cDNA encoding the catalytic domain of PKA or $CaMKII_{T286D}$ was used. Four hours after transfection, the cells were rinsed, and the media replaced. Electrophysiological recordings were performed at room temperature 1 – 2 days after transfection.

SCN8A mice

SCN8A-N1768D (C57BL/6J background) mice[65] were kindly provided by Dr. Michael Hammer (University of Arizona, Tucson, AZ). Congenic mice heterozygous for the *SCN8A*-N1768D allele were crossed with each other to generate experimental $SCN8A^{N1768D/N1768D}$ mice and wild-type $SCN8A^{+/+}$ littermate controls. Mice were kept in a vivarium on a 12-h light-dark cycle. Mice had access to regular chow and water, *ad libitum*. Both male and female mice were used. Mice were bred and procedures were conducted at the Columbia University Institute of Comparative Medicine, which is fully accredited by the Association for Assessment and Accreditation of Laboratory Animal Care. All procedures were approved by the Columbia Institutional Animal Care and Use Committee (protocol #AC-AABI1551). Intracranial injection of adeno-associated virus 9 containing ELIXIR-P2A-mCherry was performed at P0 or P1 on $SCN8A^{N1768D/N1768D}$ mice and wild-type $SCN8A^{+/+}$ mice. Pups were injected under hypothermic anesthesia. Bilateral intracerebroventricular injections of virus (1 μL/hemisphere) were performed using a Hamilton syringe to induce widespread expression in the cortex. Pups were quickly rewarmed and returned to their home cage following injection. Brains were dissected from P18-P22 animals.

$CaMKII\delta_{C,OE}$ mice

Young adult (8 to 10-wk-old, male and female) $CaMKII\delta_{C,OE}$ mice and WT littermate control mice in the C57BL6/J background were used. Mice were kept at standard temperature, humidity, and lighting. Food (Teklad, 2018) and fresh, sterile drinking water were provided *ad libitum*. Before terminal surgery, mice were injected with heparin (400 U/kg body weight) and anesthetized with isoflurane

(5% in an induction chamber, then 1.5 - 3% via nose cone). All animal handling procedures were conducted in strict compliance with a protocol (#23175) approved by the Institutional Animal Care and Use Committee at the University of California, Davis conforming to the NIH Guide for the Care and Use of Laboratory Animals (8th edition, 2011).

$Na_V1.5$ F1759A^{tg} mice

The $Na_V1.5$ F1759A mouse line was previously generated by crossing $Na_V1.5$ F1759A mice, in a B6CBA/F2 hybrid background, with αMHC-rtTA mice in a FVB/N background to generate doxycycline-inducible transgenic mice (αMHC-rtTA x tetO-$Na_V1.5$ F1759A).[2] Mice were kept on a 12-hour light-dark cycle, and fresh food and water were provided *ad libitum*. Both male and female mice were used. At P4, mouse pups were administered $1.03x10^{11}$ GC/g of adeno-associated virus 9 encoding either GFP or ELIXIR through a dorsal subcutaneous injection. ECG and electrophysiology experiments were conducted eight weeks after the injection. Prior to experimentation, the expression of $Na_V1.5$ F1759A was induced with doxycycline chow overnight. Ventricular myocytes were isolated the following day. All procedures conducted with the $Na_V1.5$ F1759A mice strictly adhered to a protocol (#AC-AABI1550) approved by Institutional Animal Care and Use Committee at Columbia University.

LQT3 cardiomyocytes from induced pluripotent stem cells

Induced pluripotent stem cells were obtained and differentiated into cardiomyocytes as described previously.[26] Prior to differentiation human iPSCs were maintained in mTeSR medium (Stem Cell Technologies) and passaged every 4 - 6 days onto Matrigel-coated plates (Corning). On the first day of differentiation (day 0), embryoid bodies (EBs) were generated by treating the iPSCs with 1 mg/mL Collagenase B (Roche, Cat #: 11088807001) for 1 h, or until cells detached from the plate. The cells were then collected and centrifuged at 300 rcf for 3 min and resuspended as small clusters of 50–100 cells by gentle pipetting in differentiation medium (RPMI 1640 (Thermo Fisher Scientific, Cat #: 11875085) with 2 mM L-glutamine (Thermo Fisher Scientific, Cat #: 25030149), 0.4 mM monothioglycerol (Millipore Sigma, Cat #: M6145), and 50 μg/mL ascorbic acid (Millipore Sigma, Cat #: A4403)). Differentiation medium was supplemented with 2 ng/mL BMP4 (R&D Systems) and 10 μM Rock inhibitor (Y-27632 dihydrochloride, Tocris Fisher Cat #: 1254/50). EBs were cultured on ultra-low attachment 6-well plates (Corning Costar, Cat #: 3471) in a humidified incubator at 37°C, 5% CO_2, and 5% O_2. On day 1, the media was changed to differentiation media supplemented with 20 ng/mL BMP4 (R&D Systems), 20 ng/mL Activin A (R&D Systems), 5ng/mL bFGF (R&D Systems). On day 3, EBs were harvested and washed once with RPMI 1640. Medium was changed to differentiation media supplemented with 5 ng/mL VEGF (R&D Systems) and 5 μM XAV939 (Reprocell-Stemgent, Cat #: 04-0046). From this point on, every 2-3 days media was replaced with media supplemented with only 5 ng/mL VEGF (R&D Systems). Following differentiation, the cells were allowed to mature for 20 – 28 days before being transduced with AAV9 encoding ELIXIR or GFP. Experiments were conducted ~5 days following transduction. Expression was visually confirmed by monitoring mCherry or GFP fluorescence.

METHOD DETAILS

Molecular biology

Standard molecular biological techniques were used to generate all new constructs (see key resources table). The human $Na_V1.5$ sequence used in this work corresponds to GenBank accession number M77235.1 and was expressed from a pGW vector, as described previously.[26] The IQ/AA mutation in the channel CTD and ΔKPQ mutation in the inactivation gate were generated using overlap-extension PCR followed by ligation into the NruI and XbaI, or KpnI and XbaI restriction sites, respectively. The Δ1864 truncation was made by PCR amplifying the relevant segment using a forward primer upstream of the KpnI site and reverse primer that truncated at position 1864, which included an XbaI site for ligation into the $Na_V1.5$/pGW construct. A gene fragment containing the DNA sequence of the designed peptide (MSPRREALYRGFRACYDVLRH) was purchased from Synbio technologies and subcloned into a pIRES2-eGFP vector using the NheI and BamHI restriction sites. For FRET experiments the Venus-tagged $Na_V1.5_{CTD}$ ($Na_V1.5$ residues 1771-2016) and Cerulean-tagged ELIXIR (WT, L8A, F12E, and Y16A) were expressed from a pCDNA3 vector. The PKA catalytic domain and $CaMKII_{T286D}$ were expressed from a pCDNA3.1 plasmid as previously reported.[92] The WT $Na_V1.1$,[93] $Na_V1.4$,[94] $Na_V1.6$,[93] and $Na_V1.7$ were expressed from pIR-CMV-IRES-mScarlet, pCDNA3.1, pCDNA4TO-IRES-mScarlet, and pCMV plasmids, respectively. For bacterial overexpression and purification, the $Na_V1.5_{CTD}$ ($Na_V1.5$ residues 1773-1940) was expressed as a GST-fusion inserted between the BamHI and XhoI sites of a pGEX-6P vector. Calmodulin was cloned into a pRSF plasmid using the NcoI and XhoI sites. QuickChange mutagenesis was used to substitute the single cystine residue (C15) with a serine to generate the final ELIXIR peptide sequence (MSPRREALYRGFRASYDVLRH), and to produce the binding-deficient ELIXIR mutants (L8A, F12E, Y16A) and the $Na_V1.6$ N1768D mutant channel. Cerulean[90], Venus,[90] C5V,[90] C32V,[90] and CTV[91] were gifts from Steven Vogel. pcDNA4TO-SCN8A-Variant-1-IRES-mScarlet and pIR-CMV-SCN1A-Variant-1-IRES-mScarlet were gifts from Al George.

Peptide design

Peptide design was done using the AfDesign algorithm implemented in ColabDesign.[31] The coordinates for the $Na_V1.5_{CTD}$ were taken from an x-ray crystallographic of the $Na_V1.5_{CTD}$ bound to calmodulin and $FHF2_U$ (PDB ID: 4DCK).[44] For the first round of design, residues 1786 - 1802 and 1850 - 1887 of $Na_V1.5$ were used as a hotspot, the binder length was set to 21 residues, and the number recycles, and models used were 1 and 2, respectively. The design protocol was then run with a semi-greedy optimizer

(pssm_semigreedy) and the default gradient decent settings. The pssm_semigreedy protocol starts by optimizing, via gradient descent, a softmax representation of the sequence. This is done for 120 steps. This results in a vector of shape binder length (21) that we call PSSM (position specific scoring matrix), which represents the probability of each amino acid at each position. For the starting sequence, the amino acid at each position is selected that maximizes the probability. For the semi-greedy protocol, at each step, 10 independent mutations are evaluated and the mutation with the best loss is fixed. The positions are sampled following 1-pLDDT[97] and the amino acid identity following PSSM. This is done for 32 steps. The sequence with best loss across the 32 steps is selected. The loss combines pLDDT over the binder positions plus contact score: 0.1 * (1-pLDDT) + 1.0 * contact score. The contact score minimizes the entropy of the distogram output (distance distribution between cβ atoms) between the target and the peptide. More specifically, every peptide position must have at least one confident distance prediction to any target position, unless a hotspot is defined. The protocol is available via Google Colab at: https://colab.research.google.com/github/sokrypton/ColabDesign/blob/v1.1.1/af/examples/peptide_binder_design.ipynb. Following the first round of design, the final peptide sequence was used as a seed to initialize the next round. This process was repeated a total of five times. The peptide sequence was then screened computationally by comparing the interface between the $Na_V1.5_{CTD}$ and the peptide in models of the complex generated with ColabFold,[41] ESMFold,[42] OmegaFold[27] and RoseTTAFold.[43] Contacts of structural units analysis was done with the COCOMAPS webserver[96] using cutoffs of 6.0 Å and 4.5 Å.

Protein expression and purification

For protein expression *Escherichia Coli* BL21 (DE+) cells were co-transformed with plasmids for both the $Na_V1.5_{CTD}$ and calmodulin (CaM) and grown in 0.5 L cultures of CIRCLEGROW supplemented with carbenicillin (0.1 mg/mL) and kanamycin (0.05 mg/mL). The cells were grown at 37 °C until OD_{600} of 0.7 was reached at which point expression was induced by the addition of 0.5 mL of 1 M IPTG and the temperature was set to 18°C. Cells were harvested the following morning by centrifugation (4,000 rpm, 4°C, 15 min) and stored at -20°C.

$Na_V1.5_{CTD}$ was purified as described previously.[98] Cell pellets were thawed and resuspended in lysis buffer (50 mM HEPES, 500 mM KCl, 50 μM EGTA, 0.01% (w/v) NaN_3, pH 7.4), disrupted via emulsification, and centrifuged (25,000 rpm, 4 °C, 50 min). The supernatant was then filtered (0.2 μM), mixed with 5 mL of Glutathione Sepharose 4B, and rocked at 22°C for 30 min. The resin was then rinsed with 100 mL of lysis buffer, and the $Na_V1.5_{CTD}$-GST+CaM complex was eluted in 15 mL of elution buffer (lysis buffer + 10 mM glutathione). The GST-tag and glutathione were removed by the addition of precision protease and overnight dialysis against lysis buffer, respectively. The sample was then mixed with 5 mL of glutathione resin and the $Na_V1.5_{CTD}$+CaM complex was collected and concentrated. Anion exchange chromatography (pH 8.5 to 7.4 and KCl 0 to 250 mM) was then used to separate the $Na_V1.5_{CTD}$ and CaM. Protein purity (≥95%) was assessed by SDS-PAGE and UV-vis spectroscopy. Protein concentration was determined from the absorbance at 280 nm.

Fluorescence anisotropy

The ELIXIR peptide, with an N-terminal fluorescein and TAT sequence (YGRKKRRQRRRGSGMSPRREALYRGFRASYDVLRH, theoretical MW: 4826.48 g/mol), was synthesized by GenScript. Binding of the $Na_V1.5_{CTD}$ to the ELIXIR peptide (100 nM) was monitored in 50 mM HEPES, 100 mM KCl, 1 mM $MgCl_2$, 100 mM EGTA, 0.01% (w/v) NaN_3, 1 mM DTT, pH 7.4 at 22 °C. The fluorescence anisotropy (λ_{em} = 494 nm, λ_{ex} = 518 nm, 5 nm excitation and emission bandpasses, 4 sec integration) was monitored throughout the titration with a fluorolog fluorometer (Horiba). Anisotropy (R) was calculated with,

$$R = \frac{I_{VV} - GI_{VH}}{I_{VV}+2GI_{VH}} \quad \text{(Equation 1)}$$

where I_{VV} and I_{VH} are the intensities of the vertically excited and vertically emitted, and vertically excited and horizontally emitted light, respectively. G was determined prior to each titration as the ratio between the horizontally excited and vertically emitted light, and the horizontally excited and horizontally emitted light (I_{HV}/I_{HH}). Each data point was the average of the three measurements.

The normalized anisotropy was fit to Equation 2 with the Solver add-in in Excel,

$$Y = \frac{K_a\,[Na_V1.5_{CTD}]_{free}}{1+K_a[Na_V1.5_{CTD}]_{free}} \quad \text{(Equation 2)}$$

where $[Na_V1.5_{CTD}]_{free}$ was determined with Equation 3

$$[Na_V1.5_{CTD}]_{free} = \frac{-b \pm \sqrt{b^2 - 4K_a\left(-[Na_V1.5_{CTD}]_{total}\right.}}{2K_a} \quad \text{(Equation 3)}$$

In Equation 3, b was $(1+K_a[\text{ELIXIR peptide}]-K_a[Na_V1.5_{CTD}]_{total})$. The reported K_D and standard deviation were determined from 3 replicate titrations.

Whole-Cell recordings from HEK293 cells

Whole-cell electrophysiology recordings were conducted at room temperature with an Axopatch 200B amplifier (Axon Instruments). Pipettes contained (in mM): 114 $CsMeSO_3$, 5 CsCl, 1 $MgCl_2$, 4 Mg-ATP, 10 HEPES (pH 7.4), and 10 BAPTA (1,2-bis(o-aminophenoxy) ethane-N,N,N′,N′-tetraacetic acid), adjusted to 290 mOsm with glucose. The bath contained (in mM) 20 NaCl, 1 $CaCl_2$, 10 HEPES (pH 7.4 adjusted with NaOH), at 300 mOsm (adjusted with glucose). Pipettes were pulled from borosilicate glass (World Precision Instruments, MTW 150-F4), with a horizontal micropipette P-97 puller (Sutter Instruments) and fire polished with a microforge (Narishige). Pipettes typically had a resistance of 2 – 3 MΩ, which was >70% compensated. An ITC-18 (Instrutech), controlled by custom MATLAB software (Mathworks), was used during data acquisition. Currents were low-pass filtered at 2 kHz before being digitized at several times that frequency. P/8 leak subtraction was used. Cells were kept at a holding potential of −120 mV.

Cell-attached Multichannel recordings

Multichannel recording from HEK293 cells, iPSC-CMs, or aMVMs were obtained in the on-cell configuration as in our previous work.[25,26] Pipettes contained a buffer comprised of (in mM): 140 NaCl, 10 HEPES, 1 $CaCl_2$, pH 7.4 (adjusted with tetraethylammonium hydroxide), and 300 mOsm (adjusted with tetraethylammonium methanesulfonate). To zero the membrane potential, the bath contained (in mM): 132 K-glutamate, 5 KCl; 5 NaCl, 3 MgCl, 2 EGTA, 20 HEPES, pH 7.4 (adjusted with NaOH), at 300 mOsm (adjusted with glucose). For experiments monitoring the dose-dependent effect of ELIXIR on I_{NaL}, HEK293 cells were treated with the indicated concentration of ELIXIR-cpp at 30 °C for 4 hours after which multichannel recordings were collected. Recordings were conducted at room temperature using the integrating mode of an Axopatch 200A amplifier (Axon Instruments, Molecular Devices). Patch pipettes were pulled from ultra-thick-walled borosilicate glass (BF200-116-10; Sutter Instruments) using a horizontal puller (P-97, Sutter Instruments), fire polished with a microforge (Narishige), and coated with Sylgard (Dow Corning). Pipettes typically had a 4–7 MΩ resistance. Elementary currents were low-pass filtered at 2 kHz with a four-pole Bessel filter and digitized at 200 kHz with an ITC-18 (Instrutech) analog-to-digital converter, controlled by custom MATLAB software (Mathworks). For every pulse P/8 leak pulses was obtained. Leak subtraction was done with an automated algorithm that fit the kinetics of the leak current and capacitive transient using convex optimization with L1 regularization of the objective function in Equation 4:

$$\min_{e,s}\left(\frac{1}{2}\left|I_L - De - s\right|_2^2 + \lambda\left|s\right|_1\right) \quad \text{s.t. } e \geq 0, s \leq 0, \text{and } s(t<t_0) = 0 \qquad \text{(Equation 4)}$$

Here, I_L is the leak waveform that is to be fit as a sum of exponentials; D is a matrix composed of a library of exponential functions with various time constants; e is a sparse vector composed of the amplitude of each exponential; s is a vector representing an estimate of negative outliers from the baseline that correspond to channel openings; and λ is a penalty term set to be 0.25. After leak subtraction, the unitary current for each patch was estimated from an amplitude histogram and each trace was subsequently idealized. The ensemble average from 50 - 120 stochastic traces was computed for each patch and normalized to the peak current. The average late current for each patch ($R_{persist}$) was computed as the average normalized P_O following 50ms of depolarization. Concentration-response data were analyzed and fit to a Hill equation using nonlinear least squares optimization (Graphpad Prism 10). The upper and lower plateau values of the curve were constrained to the R_{persit} value of $Na_V1.5$ ΔKPQ alone (upper) and co-expressed with ELIXIR (lower).

Flow cytometric FRET 2-hybrid analysis

Flow cytometric two-hybrid binding assays were performed as previously described.[95] HEK293 cells (ATCC CRL1573) cultured in 12-well plates were transfected with linear polyethylenimine 25 kDa polymer (PEI, Polysciences #2396602) using a 2:1 PEI to DNA mass ratio. For each experiment cells were transfected with 3 μg Venus-$Na_V1.5_{CTD}$, 1 μg Cerulean-ELIXIR, and 0.5 μg T antigen, after which cells were cultured for 2 days. To enhance the level of mature fluorophores protein production was inhibited with Cycloheximide, 100 μM, 2 hours before fluorescence measurements were made.

Fluorescence was measured with an LSRII (BD Biosciences) flow cytometer, equipped with 405 nm, 488 nm, and 633 nm lasers for excitation and 18 emission channels. Healthy single cells were selected based off the forward and side scatter signals. FRET efficiency was determined by measuring three different fluorescent signals: 1) Cerulean emission following direct excitation (S_{Cer}) through the BV421 channel (excitation 405 nm, emission 450/50), 2) Venus emission through direct excitation (S_{Ven}) with the fluorescein isocyanate channel (excitation 405 nm, dichroic 505 LP, emission 525/50) and 3) Venus emission due to FRET (S_{FRET}) through the BV510 channel (excitation 405 nm, dichroic 505 LP, emission 525/50). These measurements were used to obtain Cer_{direct} (Cerulean emission due to direct excitation), Ven_{direct} (Venus emission due to direct excitation) and FRET efficiency. Flow cytometric signals were collected at a medium rate (2,000 to 8,000 events/sec) and the data were exported as Flow Cytometry Standard 3.0 files for further processing and analysis with custom MATLAB software (Mathworks).

As previously described,[95] several control experiments were conducted for each experimental run of the flow cytometer. Firstly, the background fluorescence level of each channel (BG_{Cer}, BG_{Ven}, and BG_{FRET}) was determined using cells that were not exposed to any fluorophore containing plasmid. Secondly, cells expressing only Venus were used to determine the spectral crosstalk parameter R_{A1}, which corresponds to the bleed-through of Venus fluorescence into the FRET channel. Thirdly, cells expressing Cerulean alone

were used to measure the spectral crosstalk parameters R_{D1}, and R_{D2}, corresponding to the bleed-through of Cerulean fluorescence into the FRET and Venus channels, respectively. Fourth, the instrument-specific calibration parameters f_{Ven}/f_{Cer} and g_{Ven}/g_{Cer}, corresponding to the ratios of the fluorescence excitation and emission of Venus and Cerulean, respectively, were measured on the day of the experiment using Cerulean-Venus dimers with known FRET efficiencies. Fifthly, cells co-expressing Cerulean and Venus were used to estimate concentration-dependent collisional FRET. For each cell, spectral crosstalk was then accounted for with Equations 5, 6, and 7.

$$Cer_{direct} = R_{D1} \times S_{Cer} \quad \text{(Equation 5)}$$

$$Ven_{direct} = R_{A1} \times (S_{Ven} - R_{D2} \times S_{Cer}) \quad \text{(Equation 6)}$$

$$Ven_{FRET} = S_{FRET} - R_{A1} \times (S_{Ven} - R_{D2} \times S_{Cer}) - R_{D1} \times S_{Cer} \quad \text{(Equation 7)}$$

After unmixing the spectra, f_A/f_D and g_A/g_D were obtained from the Cerulean-Venus dimer data by finding the slope and intercept of the linear relationship shown in Equation 8.

$$\frac{Ven_{FRET}}{Cer_{direct}} = \frac{g_{Cer}}{g_{Ven}} \times \frac{Ven_{direct}}{Cer_{direct}} - \frac{f_{Ven}}{f_{Cer}} \quad \text{(Equation 8)}$$

Donor-centric FRET efficiencies (E_D) were then calculated with Equation 9.

$$E_D = \frac{Ven_{FRET}}{Ven_{FRET} + \frac{f_{Ven}}{f_{Cer}} \times Cer_{direct}} \quad \text{(Equation 9)}$$

The relative proportion of Cerulean (N_{Cer}) and Venus (N_{Ven}) fluorophores in each cell were then determined from Equations 10 and 11, respectively.

$$N_{Cer} = \frac{Cer_{direct}}{1 - E_D} \quad \text{(Equation 10)}$$

$$N_{Ven} = \frac{Ven_{direct}}{\left(\frac{g_{Ven}}{g_{Cer}} - \frac{f_{Ven}}{f_{Cer}}\right)} \quad \text{(Equation 11)}$$

A 1:1 binding isotherm was then imposed by iteratively fitting the data to Equation 12,

$$E_D = E_{Max} \frac{Ven_{Free}}{Ven_{Free} + K_{D,EFF}} \quad \text{(Equation 12)}$$

where Ven_{Free} is determined with Equation 13,

$$Ven_{Free} = N_{Ven} - (N_{Cer} - Cer_{Free}) \quad \text{(Equation 13)}$$

in which Cer_{Free} is computed from Equation 14.

$$Cer_{Free} = 0.5(N_{Cer} - K_{D,EFF} - N_{Ven}) + 0.5\left(\sqrt{(N_{Cer} - K_{D,EFF} - N_{Ven})^2 + 4N_{Cer}K_{D,EFF}}\right) \quad \text{(Equation 14)}$$

For each FRET pair, $K_{D,EFF}$, E_{Max}, and 95% confidence intervals were obtained through iterative constrained least squares analysis. Data were obtained from two independent transfections.

Phylogenetic tree construction

The sequences of the nine human Na_V CTDs were obtained from UniProt.[99] The following residues were used for each isoform: $Na_V1.1$: 1787-1954; $Na_V1.2$: 1777-1944; $Na_V1.3$: 1772-1939; $Na_V1.4$: 1599-1776; $Na_V1.5$: 1773-1940; $Na_V1.6$: 1767-1934; $Na_V1.7$: 1761-1928; $Na_V1.8$: 1723-1890; $Na_V1.9$: 1605-1772. The sequences were aligned with the BlastP algorithm and the phylogenetic tree constructed using the BLAST server.[100]

Adeno-associated virus generation

Adeno-associated virus 9 for the expression of ELIXIR, and control virus for the expression of GFP, were purchased from VectorBuilder. To monitor the expression of ELIXIR it was cloned with a C-terminal P2A site followed by mCherry (ELIXIR-P2A-mCherry).

Flow cytometric analysis of ELIXIR uptake

Uptake of the ELIXIR-cpp was monitored using the same peptide as used in the fluorescence anisotropy experiments. Freshly dissociated aMVM were plated in a 12 well plate and incubated with different concentrations (100 nM – 10 μM) of ELIXIR-cpp for 2 hours at 37 °C in Tyrode's solution (135 mM NaCl, 4 mM KCl, 1 mM $MgCl_2$, 10 mM HEPES, 1 mM $CaCl_2$ (pH 7.4 using NaOH)). Cellular uptake was monitored with an LSR II (BD Biosciences) flow cytometer equipped with 405 nm, 488 nm, and 633 nm lasers for excitation and 18 different emission channels as previously described.[26] Forward and side scatter signals were detected and used to gate for single cells that were living. Flow cytometric signals were collected at a medium flow rate (2k - 8k events/sec). Histograms corresponding to the single-cell population of the fluorescein channel were analyzed using custom MATLAB software (Mathworks).

Brain slice electrophysiology

Slice electrophysiology was conducted at P18-P22, as homozygous expression of the *SCN8A*-N1768D allele causes early mortality. Animals were deeply anesthetized with isofluorane and decapitated, and the brain was quickly dissected into ice-cold high-sucrose slicing solution (26 mM NaHCO3, 2.5 mM KCl, 1.25 mM NaH2PO4, 10 mM MgCl2, and 0.5 mM CaCl2, 11 mM d-glucose, 234 mM sucrose, pH 7.3-7.4). 250 μm slices were cut using a Leica VT1500 Vibratome. Slices were transferred to 37°C aCSF (126 mM NaCl, 26 mM NaHCO3, 2.5 mM KCl, 1.25 mM NaH2PO4, 1.0 mM MgCl2, and 2.0 mM CaCl2, 10 mM d-glucose, pH 7.3-7.4) and incubated for 1 hour. All solutions were continuously bubbled with 95% O2 and 5% CO2. Individual slices were transferred to a recording chamber located on an upright microscope (BX51WI; Olympus) and were perfused with heated (32 °C) oxygenated aCSF.

Whole-cell current-clamp recordings were obtained using Multiclamp 700B and Clampex 10.7 software (Molecular Devices). Layer V pyramidal cells were targeted for patch-clamp experiments based on location, size, and shape. Cells expressing ELIXIR-P2A-mCherry were identified using a fluorescent light source (Olympus). Both ELIXIR-P2A-mCherry expressing and non-expressing cells were targeted for patch clamp to generate within-subject control data. Intracellular solution contained 127 mM K-gluconate, 10 mM HEPES, 8 mM NaCl, 0.6 mM EGTA, 4 mM ATP, and 0.3 mM GTP, pH 7.2. When patch electrodes were filled with intracellular solution, their resistance ranged from 3.5 to 6 MΩ. Access resistance was monitored continuously for each cell and any cell that changed by more than 15% was excluded from analyses.

Holding current was delivered to adjust resting membrane potential to -75 mV. AP-generating properties of neurons were tested by injecting 1 sec square pulses of depolarizing current incrementing in 20 pA steps, up to 1 nA. AP threshold was defined as the membrane voltage at which the slope crossed 5 mV/ms. APs were automatically identified using this slope threshold and visually confirmed. APD_{90} was defined as the time during which the membrane voltage exceeded 10% of the peak amplitude. Half-width was defined as the time during which the membrane voltage exceeded 50% of the peak amplitude. Maximum rise rate was defined as the maximum of the first derivative of the rising phase of the AP. Maximum fall rate was defined as the minimum of the first derivative of the falling phase of the AP. Depolarization block threshold was defined as the current step at which the number of APs produced fell to less than ½ of the maximum. Brain slice electrophysiology data was analyzed offline with custom MATLAB software.

Enzymatic isolation of ventricular CaMKII$\delta_{C,OE}$ cardiomyocytes

Hearts from CaMKII$\delta_{C,OE}$ mice and WT littermate were excised and retrograde perfused on a constant flow Langendorff apparatus (4 min, 37 ºC) with Ca^{2+}-free normal Tyrode's solution, gassed with 100% O_2. The heart was then perfused for 12 - 17 min with 100 mg collagenase (type 2, Worthington Biochemical Corp) and 1.4 mg protease (type XIV, Sigma-Aldrich) in 50 mL of Tyrode's solution (with 10 μmol/L Ca^{2+}) to enzymatically isolate cardiomyocytes. Following digestion, the myocytes were gently triturated with a pipette, then filtered through a nylon mesh and allowed to sediment for ~ 10 min. The sedimentation was repeated three times with increasing [Ca^{2+}] from 0.125 to 0.25 to 0.5 mmol/L. Finally, ventricular myocytes were kept in Tyrode's solution (0.5 mmol/L Ca^{2+}) at room temperature until use.

CaMKII$\delta_{C,OE}$ cardiomyocyte electrophysiology

Isolated single murine ventricular cardiomyocytes were placed in a temperature-controlled perfusion chamber (Warner Instruments) and mounted on a Leica DMI3000 B inverted microscope (Leica Microsystems). Cells were bathed at 37°C, for 10 minutes, before starting, and continuously perfused (2 mL/min) during experiments with Tyrode's solution containing (in mmol/L): NaCl 140, KCl 4, $CaCl_2$ 1.8, $MgCl_2$ 1, HEPES 5, Na-HEPES 5, glucose 5.5; pH=7.4. Electrodes were fabricated from borosilicate glass (World Precision Instruments) and had tip resistances of 2 to 3 MΩ when filled with internal solution which contained (in mmol/L): K-aspartate 100, KCl 30, NaCl 8, Mg-ATP 5, phosphocreatine dipotassium salt 10, HEPES 10, EGTA 0.01, cAMP 0.002, and calmodulin 0.0001; pH=7.2 (with KOH). Using this internal solution, the intracellular Ca^{2+} transient and contraction of the cardiomyocyte are preserved. An Axopatch 200B amplifier (Axon Instruments Inc.) was used for recording and signals were digitized at 50 kHz by a Digidata 1322A A/D

converter (Axon Instruments) under software control (pClamp10.4). Series resistance was typically 3 to 5 MΩ and was compensated by ≥90%. Experiments were discarded when the series resistance was high or increased by ≥20% during the recordings. All experiments were conducted at 37±0.1°C.

APs were recorded in whole-cell current-clamp conditions where cells were stimulated with supra-threshold depolarizing pulses (2 ms duration) delivered via the patch pipette at 1 Hz. AP duration at 90% repolarization (APD_{90}) was used to characterize AP repolarization. Series of 50 consecutive APs were analyzed to estimate short-term variability of the APD_{90}. Short-term variability was calculated with Equation 5,

$$STV = \frac{\sum(|APD_{n+1} - APD_n|)}{(n_{beats} - 1) \times \sqrt{2}} \quad \text{(Equation 15)}$$

where APD_n and APD_{n+1} indicate the durations of the nth and $(n+1)^{th}$ AP, and n_{beats} denotes the total number of consecutive beats analyzed.

For voltage-clamp late Na^+ current measurements, internal solution contained (in mmol/L): CsCl 110, tetraethylammonium chloride 20, Mg-ATP 5, HEPES 10, phosphocreatine disodium salt 5, calmodulin 0.0001, EGTA 10, $CaCl_2$ 4.1 (free [Ca^{2+}] =100 nmol/L), pH=7.20. Bath solution contained (in mmol/L): NaCl 140, CsCl 4, $CaCl_2$ 1.8, $MgCl_2$ 1, HEPES 5, Na-HEPES 5, glucose 5.5, 4-aminopyridine 5, nifedipine 0.01, pH=7.4. The late Na^+ current magnitude was measured at the end of a 500 ms depolarizing pulse from a holding potential of -120 mV to -40 mV. The current amplitude was normalized to the capacitance (late Na^+ current density) of each cell. Chemicals and reagents were purchased from Sigma-Aldrich. A subset of cells was preincubated with ELIXIR-cpp (3 μmol/L, 4 hours).

Confocal imagining

Images of freshly dissociated myocytes were collected on a Nikon Ti Eclipse inverted microscope equipped with a Yokogawa CSU-X1 spinning disk confocal and a CMOS camera from Andor Zyla scientific using a 60x oil-immersion objective.

ECG analysis

Six-lead electrocardiograms were recorded from isoflurane-anesthetized mice using an Emka ECG with subcutaneous electrodes and IOX software (Emka Technologies). The RR QRS and QT durations were measured manually with IOX software. The QT duration was corrected for heart rate with Mitchell's formula[74] (Equation 16):

$$QT_C = \frac{QT}{\sqrt{\frac{RR}{100}}} \quad \text{(Equation 16)}$$

where QT and RR are the measured QT and RR interval, respectively.

Images of protein structures

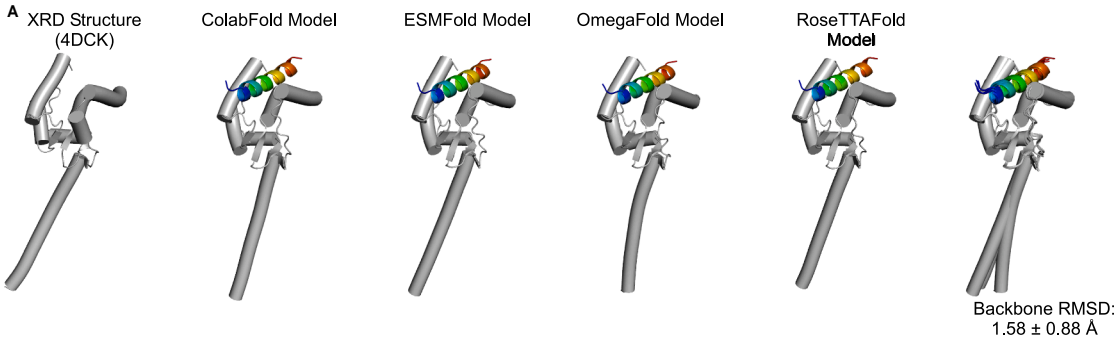
All protein structure images and vacuum electrostatics were generated with PyMOL (Schrödinger LLC). Unless otherwise stated, the $Na_V1.5_{CTD}$ is shown as gray cylinders and the ELIXIR peptide as a light blue (actinium) helix.

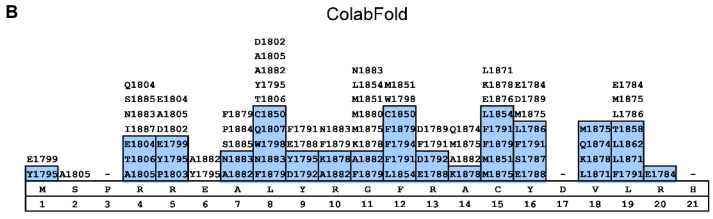
QUANTIFICATION AND STATISTICAL ANALYSIS

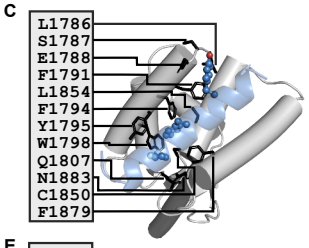
Pooled data are presented as the mean ± SEM. Statistical analysis was performed in Graphpad Prism 10.0.0. Information about statistical analysis is provided in figures and respective figure legends. Normality was tested using a D'Agostino-Pearson normality test. For normally distributed data requiring multiple comparisons, a one-way analysis of variance (ANOVA) followed by Dunnett's multiple comparisons was used. Multiple comparisons of nonnormally distributed data were done with a Kruskal-Wallis test followed by a Dunn's post hoc test. Comparisons of two groups were conducted with a 2-tailed Student's t test (normally distributed data), or a Mann-Whitney U test (nonnormally distributed data). Differences with a p value < 0.05 were considered statistically significant. For recordings conducted on aMVMs, the normality of the data was assessed by Shapiro-Wilk test and the equality of group variance was tested using the Brown-Forsythe test. Statistical significance of differences was determined using Mann-Whitney U test, and ANOVA with an appropriate multiple comparisons test, when applicable. Interaction between genotype and treatment was determined using a two-way ANOVA. Animals were grouped with no blinding but randomized in cellular experiments. Fully blinded analysis was not performed in cellular studies because the same person carried out the experiments and analysis. Group sizes were determined by a priori power analysis with $\alpha = 0.05$ and power of 0.8, to detect a 20% signal difference at the endpoint.

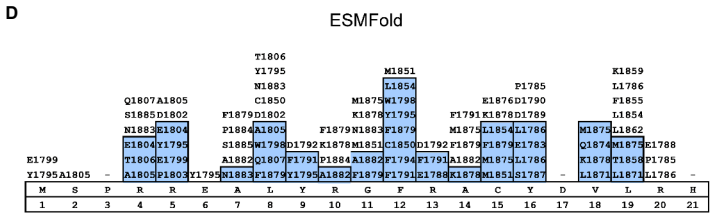
CellPress

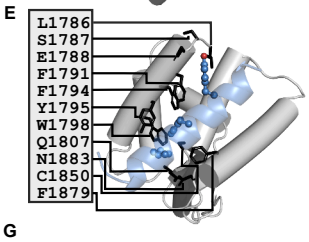
Supplemental figures

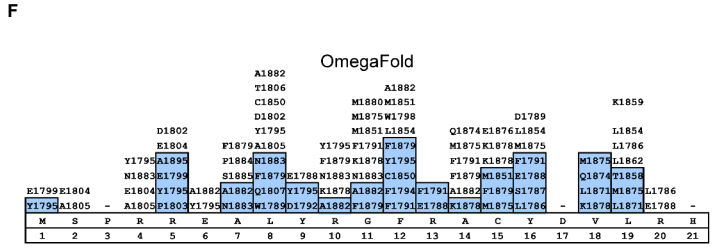


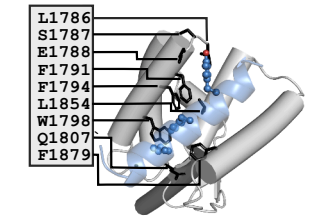


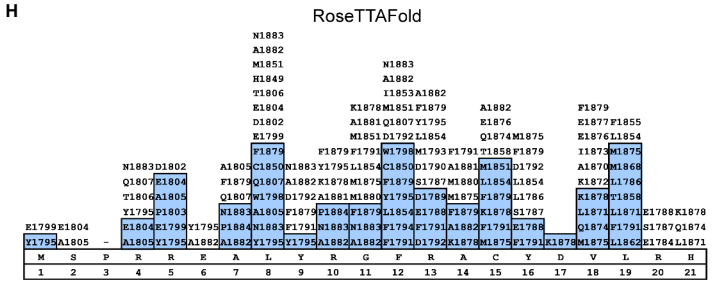


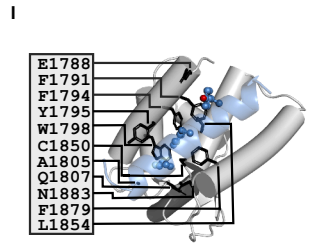


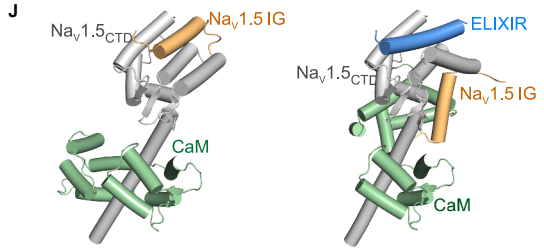












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Figure S1. Models of the ELIXIR-Na$_V$1.5 CTD complex, related to Figure 1

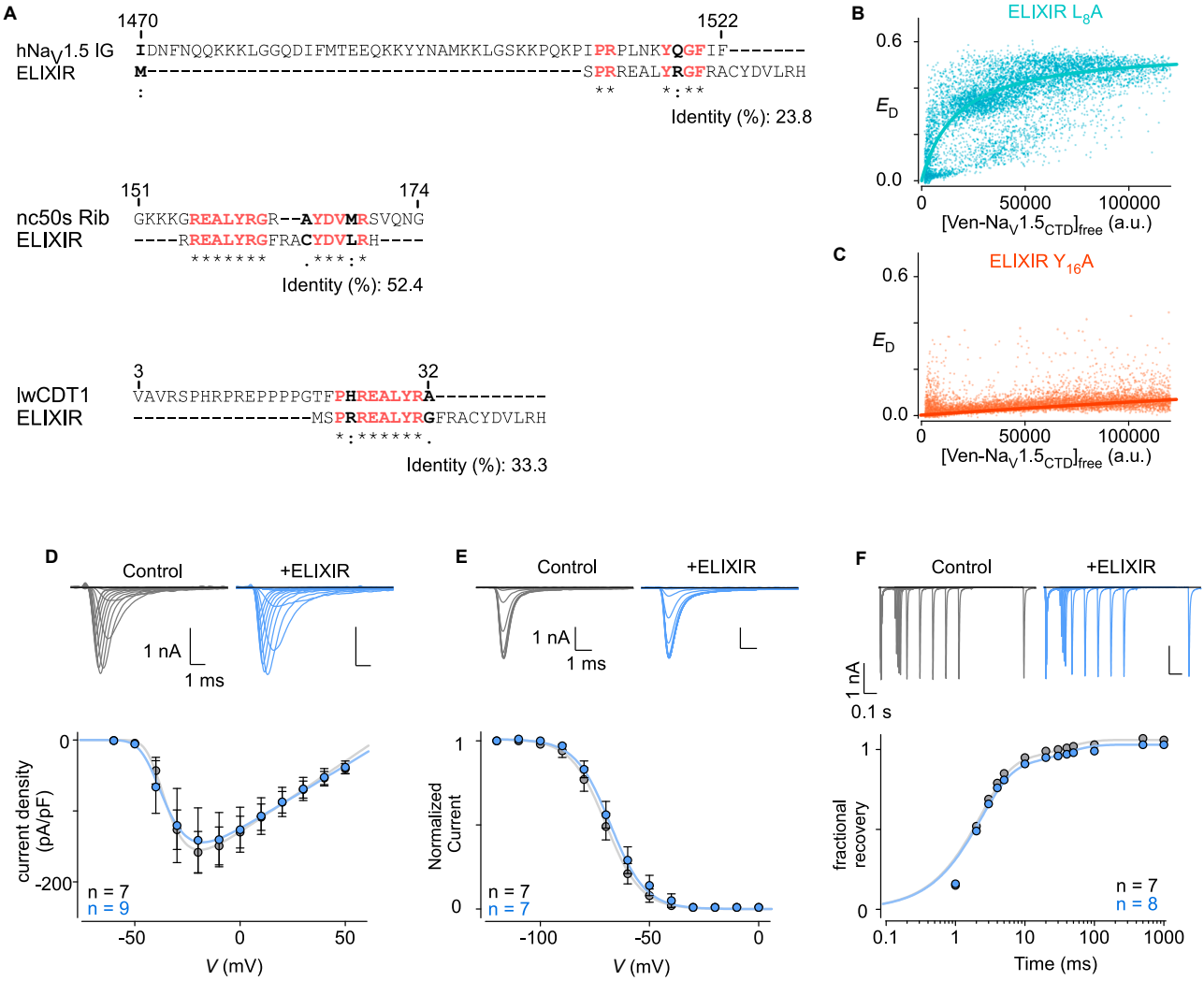
(A) Comparison of a high-resolution structure of the Na$_V$1.5 CTD (PDB: 4DCK) to computational models of the Na$_V$1.5 CTD-ELIXIR complex generated with ColabFold, ESMFold, OmegaFold, and RoseTTAFold. The Na$_V$1.5 CTD is shown as gray cylinders, and the ELIXIR peptide is shown in rainbow (blue, N terminus; red, C terminus). The models were aligned using residues 1,786–1,892 of the model generated with ColabFold.

(B) Contacts of structural units analysis of the ELIXIR-Na$_V$1.5 CTD models generated with ColabFold. Na$_V$1.5 residues within 6 Å (within 4.5 Å colored blue) of the ELIXIR peptide are listed.

(C) Residues of the Na$_V$1.5 CTD (black sticks) within 6 Å of the ELIXIR peptide shown in the ELIXIR-Na$_V$1.5 CTD complexes generated with ColabFold. The Na$_V$1.5 CTD is shown as gray cylinders, and ELIXIR as a light blue (actinium) α helix. Models were aligned as in (A).

(D–I) Contacts of structural units analysis of the ELIXIR-Na$_V$1.5 CTD models with ESMFold (D), OmegaFold (F), and RoseTTAFold (H). Residues of the Na$_V$1.5 CTD (black sticks) within 6 Å of the ELIXIR peptide shown in the ELIXIR-Na$_V$1.5 CTD complexes generated with ESMFold (E) OmegaFold (G), and RoseTTAFold (I).

(J) ColabFold generated models of the Na$_V$1.5 CTD + Na$_V$1.5 IG + calmodulin (CaM) complex without (left) and with (right) ELIXIR. CaM is shown as pale green cylinders, and the Na$_V$1.5 IG as a light orange cylinder. The Na$_V$1.5 CTD and ELIXIR are colored as in (C). Models were aligned as in (A).



G

	$V_{1/2}$ (mV)	V_{rev} (mV)	Slope Factor	g
WT $Na_V1.5$	-33.59 ± 1.52	69.48 ± 1.61	4.30 ± 0.53	33.83 ± 5.91
WT $Na_V1.5$ + ELIXIR	-29.26 ± 2.76	69.51 ± 1.21	4.85 ± 0.39	36.20 ± 6.25

Figure S2. ELIXIR sequence alignments and whole-cell electrophysiology, related to Figure 1

(A) Alignment of the human $Na_V1.5$ IG (top, h$Na_V1.5$ IG), *Nannochloropsis gaditana* 50s ribosome (middle, nc50s Rib), and *Leptonychotes weddellii* DNA replication factor CDT1 (bottom, lwCDT1) sequences with ELIXIR. Similar residues are shown in bold, and identical residues are colored red and shown in bold.

(B and C) FRET efficiency between cerulean-tagged ELIXIR L8A (B) or ELIXIR Y16A (C) and Venus-tagged $Na_V1.5_{CTD}$ plotted against the concentration of free Venus-tagged $Na_V1.5_{CTD}$, a.u. is arbitrary units. The solid lines depict the fit of a 1:1-binding isotherm.

(D) Peak current density of WT $Na_V1.5$ in the absence (gray) or presence of ELIXIR (light blue). Data are shown as the mean ± SEM, with the number of cells (*n*) listed in parentheses. Exemplar currents for WT $Na_V1.5$ alone (gray) and with ELIXIR (light blue) are shown above the plot.

(E) Steady-state inactivation relationships for WT $Na_V1.5$ in the absence (gray) and presence of ELIXIR (light blue). Data are shown as the mean ± SEM, with the number of cells (*n*) shown in parentheses. Exemplar currents for WT $Na_V1.5$ alone (gray) and with ELIXIR (light blue) are shown above the plot.

(F) Fractional recovery of WT $Na_V1.5$ from inactivation in the absence (gray) and presence of ELIXIR (light blue). Data are shown as the mean ± SEM, with the number of cells (*n*) listed. Exemplar currents for WT $Na_V1.5$ alone (gray) and with ELIXIR (light blue) are shown above the plot.

(G) Summary of fit parameters from whole-cell recordings shows minimal changes in voltage-dependence of channel activation (mean ± SEM from 7 and 9 cells, without and with ELIXIR, respectively).

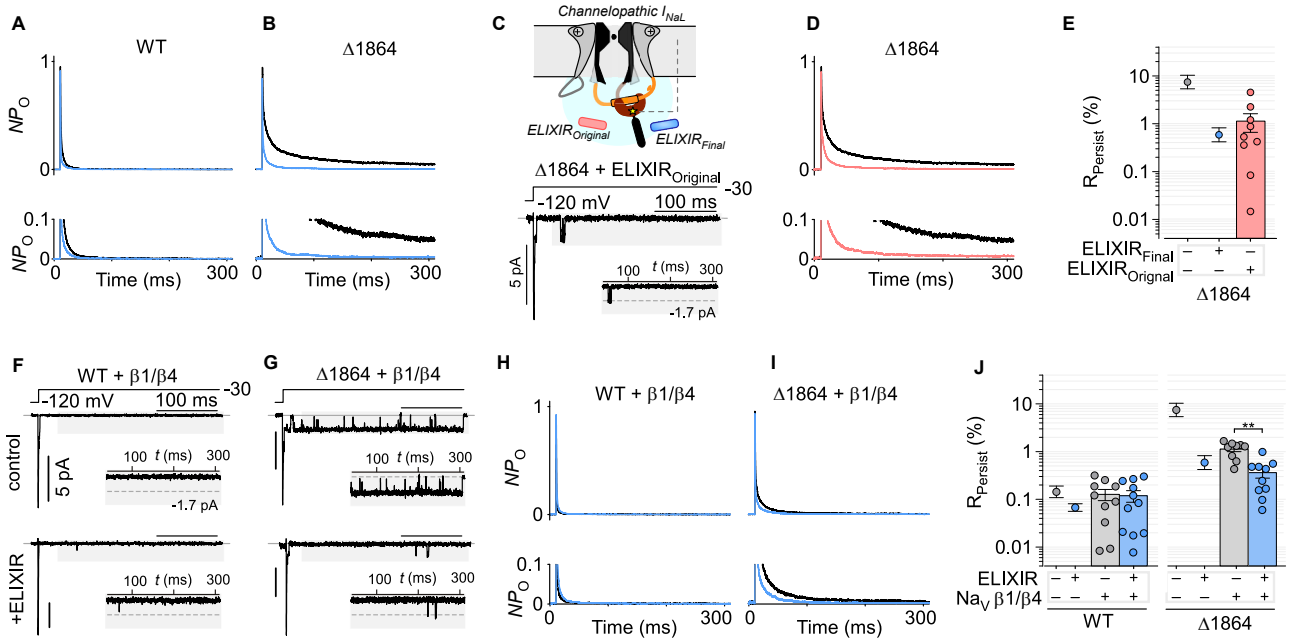


Figure S3. Effect of Na_V β-subunits and mutations on ELIXIR, related to Figure 2

(A and B) Comparison of the ensemble average NP_o of $Na_V1.5$ WT (A) and Δ1864 (B) in the absence (black) or presence (light blue) of ELIXIR. The lower panel shows an expanded view near the baseline.

(C) Schematic of $Na_V1.5$ in the open state with $ELIXIR_{Original}$ (light red) and $ELIXIR_{final}$ (light blue) (top). Exemplar multichannel recording of $Na_V1.5$ Δ1864 co-expressed with the initial ELIXIR peptide sequence ($ELIXIR_{Original}$; MSPRREALYRGFRACYDVLRH) (bottom). Inset shows the late phase enlarged for better visualization.

(D) Comparison of the ensemble average NP_o of $Na_V1.5$ Δ1864 in the absence (black) and presence of $ELIXIR_{Original}$ (light red). The lower panel shows an expanded view near the baseline.

(E) Bar graph summarizing the effect of $ELIXIR_{Original}$ on the I_{NaL} of $Na_V1.5$ Δ1864 (n = 9 cells, 632 sweeps). Data are represented as mean ± SEM. The gray and light blue circles show the mean ± SEM $R_{persist}$ of the channel alone or with ELIXIR, respectively.

(F and G) Exemplar multichannel recordings of $Na_V1.5$ WT (F) and Δ1864 (G) co-expressed with β1 and β4 in the absence (top) or presence (bottom) of ELIXIR. Inset shows the late phase enlarged for better visualization.

(H and I) Comparison of the ensemble average NP_o of $Na_V1.5$ WT (H) or Δ1864 (I) co-expressed with β1 and β4 in the absence (black) or presence (light blue) of ELIXIR. The lower panel shows an expanded view near the baseline.

(J) Bar graph summarizes effect of β1/β4-subunits and ELIXIR on I_{NaL} of $Na_V1.5$ WT and Δ1864. Each bar, mean ± SEM. Data in the absence of β1/β4 is shown as dot with error bar denoting mean ± SEM. $Na_V1.5$ with β1/β4 alone (n = 10 cells, 953 sweeps) or with β1/β4 and ELIXIR (n = 11 cells; 921 sweeps), Na_V Δ1864 with β1/β4 alone (n = 9 cells, 775 sweeps) or with β1/β4 and ELIXIR (n = 10 cells; 960 sweeps). $^{**}p$ = 0.0021 ($Na_V1.5$ Δ1864 + β1/β4 + ELIXIR) compared with $Na_V1.5$ Δ1,864 + β1/β4 by a Brown-Forsythe test followed by Dunn's multiple comparison test.

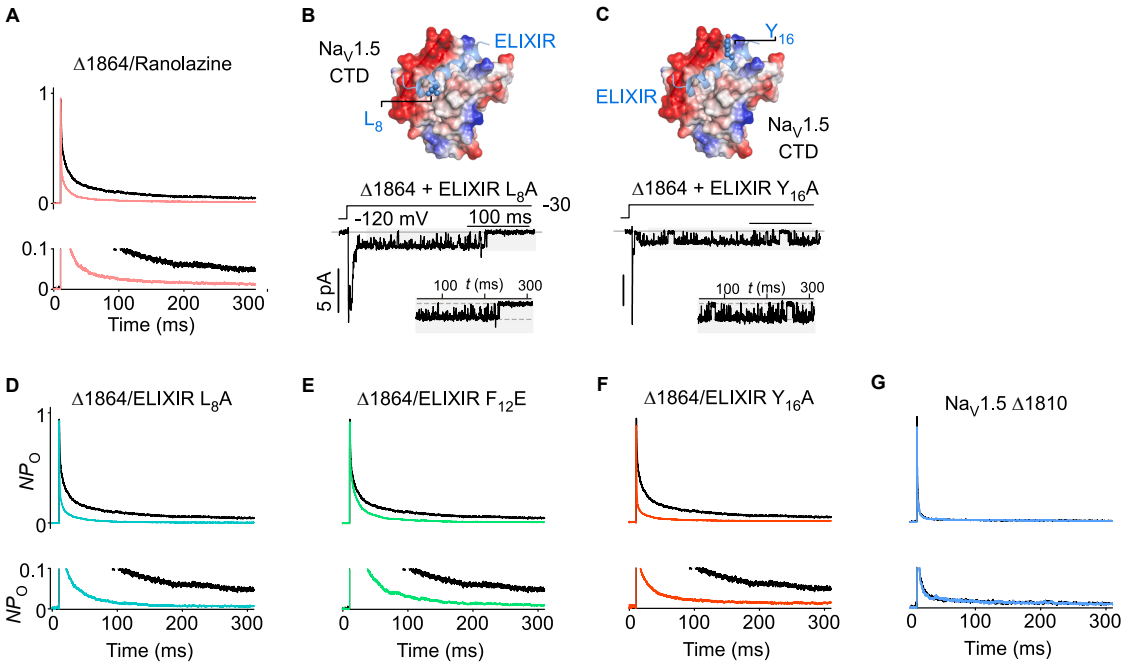


Figure S4. Mutations in ELIXIR reduce I_{NaL} inhibition, related to Figure 2

(A) Comparison of the ensemble average NP_o of Na$_V$1.5 Δ1864 in the absence (black) and presence (pink) of 10 μM ranolazine. The lower panel shows an expanded view near the baseline.

(B and C) Top: vacuum electrostatic surface (positive: blue, neutral, and white; negative: red) of the Na$_V$1.5 CTD EFL bound to ELIXIR with L8 or Y16 shown as ball and stick (top). Bottom: exemplar multichannel recordings of Na$_V$1.5 Δ1864 in the presence of ELIXIR L8A or ELIXIR Y16A.

(D–F) Comparison of the ensemble average NP_o of Na$_V$1.5 Δ1864 mutant with ELIXIR mutants L8A (D), F12E (E), and Y16A (F). Baseline, black, and with ELIXIR are shown in color.

(G) Comparison of the ensemble average NP_o of Na$_V$1.5 Δ1810 mutant with ELIXIR. Baseline, black, and with ELIXIR (light blue).

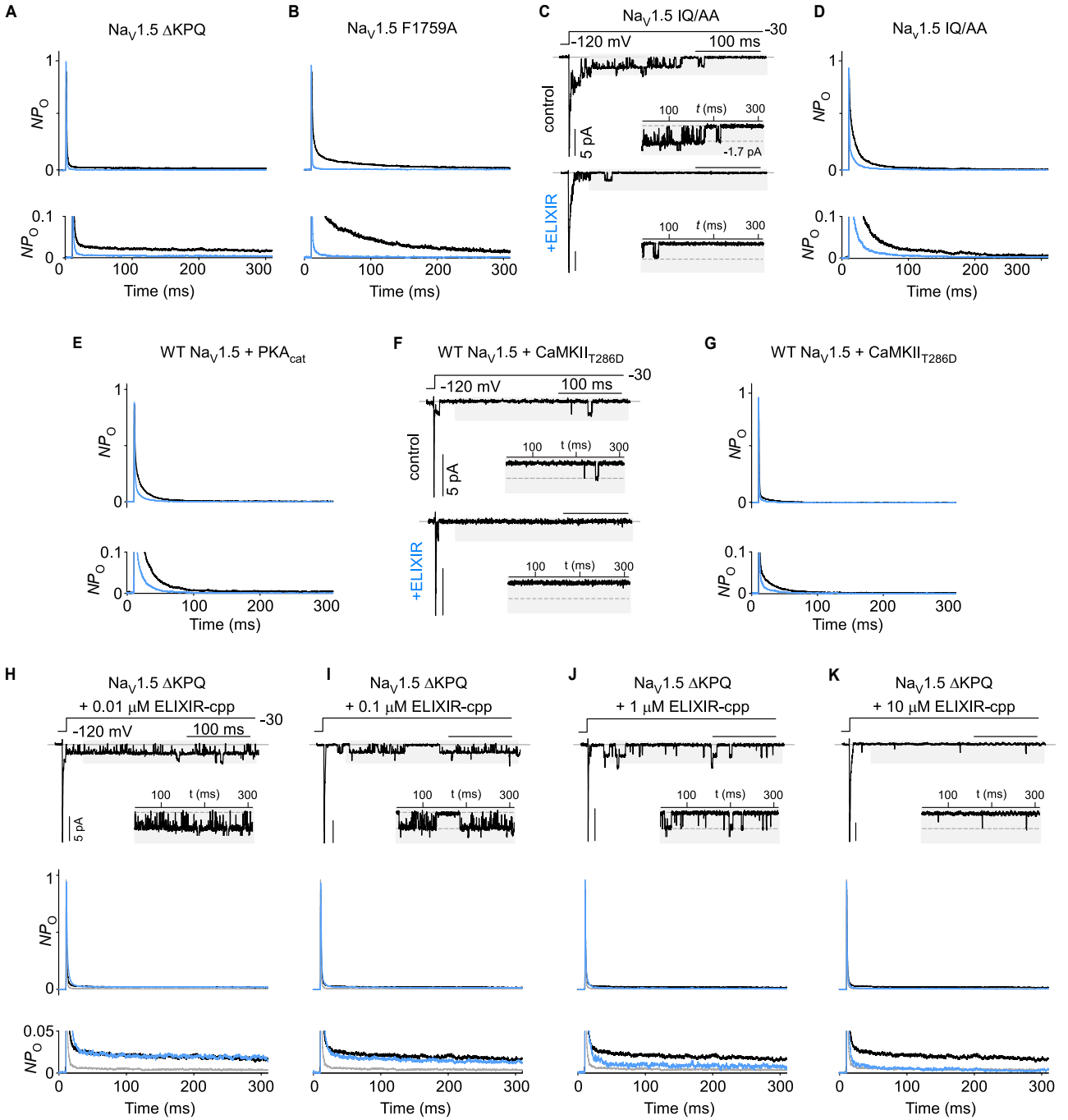


Figure S5. Generality of I_{NaL} inhibition by ELIXIR, related to Figure 3

(A and B) Comparison of the ensemble average NP$_o$ of Na$_V$1.5 ΔKPQ (A) and Na$_V$1.5 F1759A (B) in the absence (black) and presence of ELIXIR (light blue). The lower panel shows an expanded view near the baseline.

(C) Exemplar multichannel recordings of Na$_V$1.5 IQ/AA in the absence (top) and presence (bottom) of ELIXIR. Inset shows the late phase enlarged for better visualization.

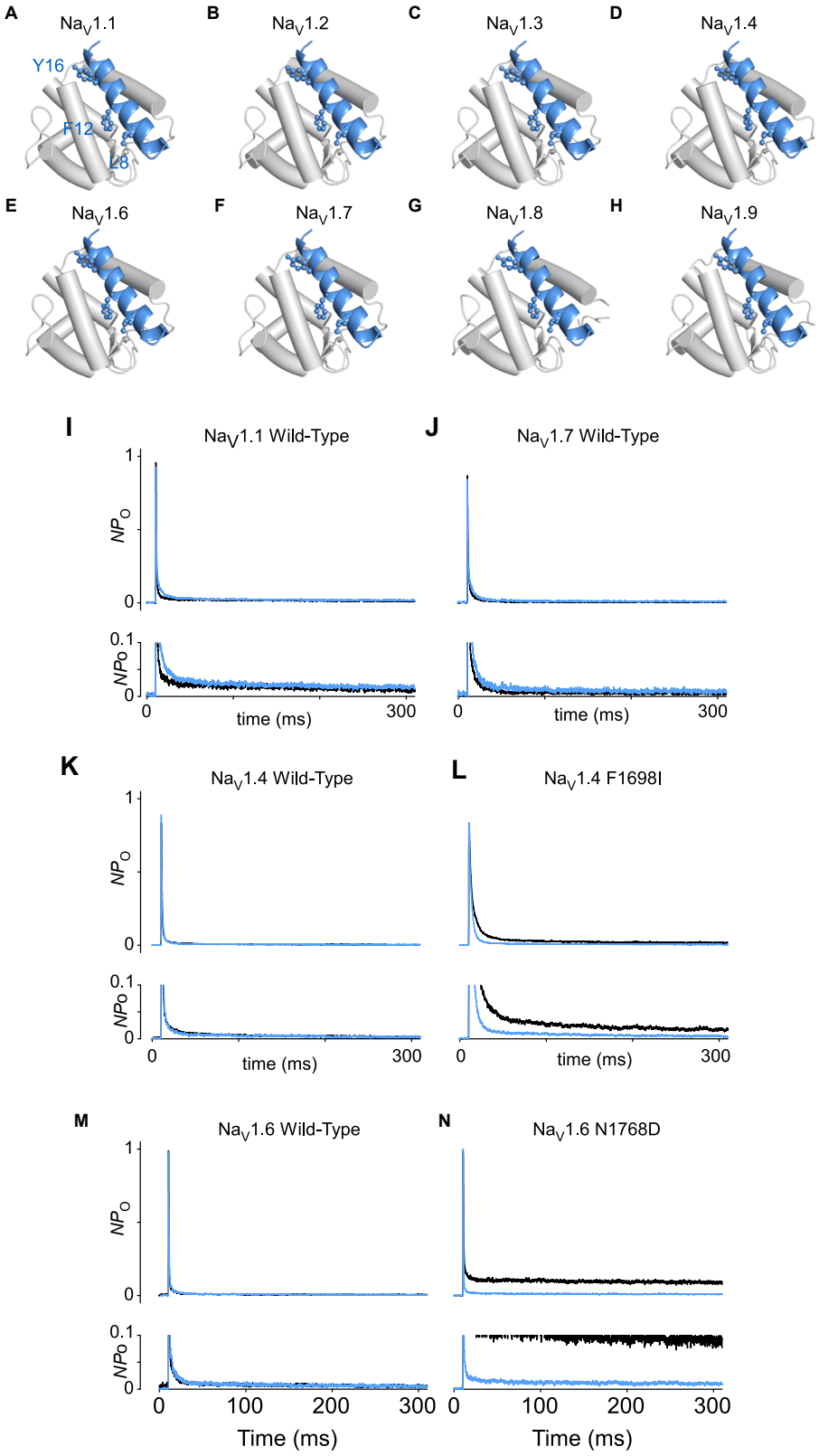
(D and E) Comparison of the ensemble average NP$_o$ of Na$_V$1.5 IQ/AA (D) and WT Na$_V$1.5 + PKA$_{Cat}$ (E) in the absence (black) and presence of ELIXIR (light blue). The lower panel shows an expanded view near the baseline.

(F) Exemplar multichannel recordings of WT Na$_V$1.5 + CaMKII$_{T286D}$ in the absence (top) and presence (bottom) of ELIXIR. Inset shows the late phase enlarged for better visualization.

(G) Comparison of the ensemble average NP$_o$ of WT Na$_V$1.5 + CaMKII$_{T286D}$ in the absence (black) and presence of ELIXIR (light blue). The lower panel shows an expanded view near the baseline.

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(H–K) Exemplar multichannel recordings (top) and comparison of the ensemble average NP_O (bottom) of $Na_V1.5$ ΔKPQ following treatment with 0.01 μM ELIXIR-cpp (H), 0.1 μM ELIXIR-cpp (I), 1 μM ELIXIR-cpp (J), and 10 μM ELIXIR-cpp (K). In each panel, the ensemble average NP_o of $Na_V1.5$ ΔKPQ in the absence of ELIXIR is shown in black, with ELIXIR overexpressed in gray and following treatment with the designated concentration of ELIXIR-cpp in light blue. In the upper panel, the insets show the late phase enlarged for better visualization. The bottom panel shows an expanded view, near the baseline, of the ensemble average NP_O.



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Figure S6. ColabFold models of ELIXIR-Na_V CTD complexes, related to Figure 4

(A–H) ColabFold generated models of ELIXIR (light blue) bound to the CTD (gray cylinders) of $Na_V1.1$ (A), $Na_V1.2$ (B), $Na_V1.3$ (C), $Na_V1.4$ (D), $Na_V1.6$ (E), $Na_V1.7$ (F), $Na_V1.8$ (G), and $Na_V1.9$ (H). ELIXIR residues L8, F12, and Y16 are shown as ball and stick. The structures were aligned to residues 1,788–1,893 of $Na_V1.5$. For clarity, only the EFL region (residues 1,788–1,893 of $Na_V1.5$) of each CTD is shown.

(I–N) Comparison of the ensemble average NP_o of WT $Na_V1.1$ (I), WT $Na_V1.7$ (J), WT $Na_V1.4$ (K), $Na_V1.4$ F1698I (L), WT $Na_V1.6$ (M), and $Na_V1.6$ N1768D (N) in the absence (black) and presence of ELIXIR (light blue). The lower panel shows an expanded view near the baseline.

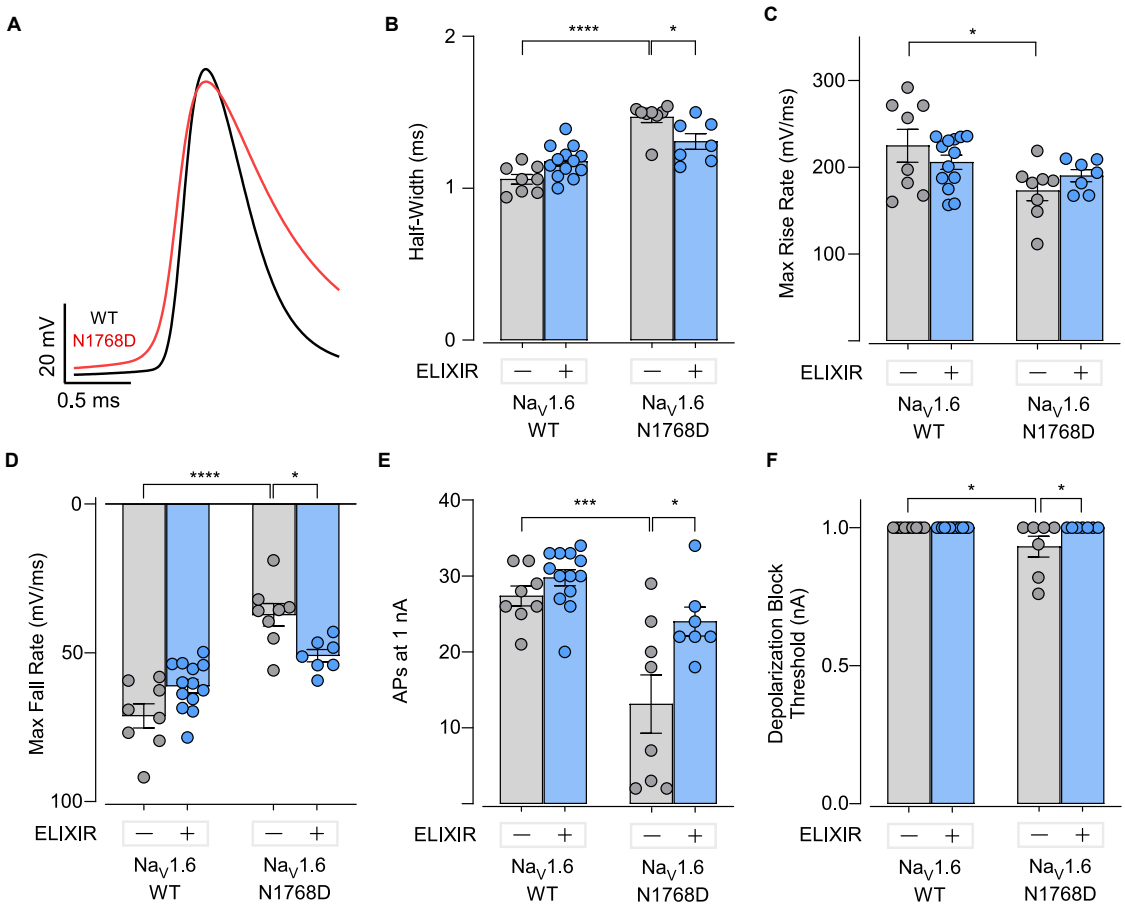


Figure S7. Effect of ELIXIR on WT Na$_V$1.6 and Na$_V$1.6 N1768D neuronal AP properties, related to Figure 5

(A) Exemplar AP recordings of control neurons from WT Na$_V$1.6 (black) and Na$_V$1.6 N1768D (red) mice.

(B) Bar graph summarizing the effect of ELIXIR on the AP half-width of control and ELIXIR-expressing neurons from WT Na$_V$1.6 (control $n = 8$ cells from 3 mice; ELIXIR $n = 13$ cells from 3 mice) or Na$_V$1.6 N1768D mice (control $n = 8$ cells from 2 mice; ELIXIR $n = 7$ cells from 2 mice). Each bar, mean ± SEM. ****$p < 0.0001$ (WT Na$_V$1.6 control), *$p = 0.0330$ (Na$_V$1.6 N1768D ELIXIR) compared with Na$_V$1.6 N1768D control by a two-way ANOVA followed by a Tukey's multiple comparisons test.

(C) Bar graph summarizing the effect of ELIXIR on the maximum rise rate of APs from control and ELIXIR-expressing neurons from WT Na$_V$1.6 (control $n = 8$ cells from 3 mice; ELIXIR $n = 13$ cells from 3 mice) and Na$_V$1.6 N1768D (control $n = 8$ cells from 2 mice; ELIXIR $n = 7$ from 2 mice) mice. Each bar, mean ± SEM. *$p = 0.0296$ (WT Na$_V$1.6 control) compared with Na$_V$1.6 N1768D control by a two-way ANOVA followed by a Tukey's multiple comparisons test.

(D) Bar graph summarizing the effect of ELIXIR on the maximum fall rate of APs from control and ELIXIR-expressing neurons from WT Na$_V$1.6 (control $n = 8$, $N = 3$ mice; ELIXIR $n = 13$, $N = 3$ mice) and Na$_V$1.6 N1768D (control $n = 8$ cells from 2 mice; ELIXIR $n = 7$ cells from 2 mice) mice. Each bar, mean ± SEM. ****$p < 0.0001$ (WT Na$_V$1.6 control), *$p = 0.0320$ (Na$_V$1.6 N1768D ELIXIR) compared with Na$_V$1.6 N1768D control by a two-way ANOVA followed by a Tukey's multiple comparisons test.

(E) Bar graph summarizing the effect of ELIXIR on the number of APs fired at 1 nA from control and ELIXIR-expressing neurons from WT Na$_V$1.6 (control $n = 8$ cells from 3 mice; ELIXIR $n = 13$ cells from 3 mice) and Na$_V$1.6 N1768D mice (control $n = 8$ cells from 2 mice; ELIXIR $n = 7$ cells from 2 mice). Each bar, mean ± SEM. ***$p < 0.0004$ (WT Na$_V$1.6 control), *$p = 0.0104$ (Na$_V$1.6 N1768D ELIXIR) compared with Na$_V$1.6 N1768D control by a two-way ANOVA followed by a Tukey's multiple comparisons test.

(F) Bar graph summarizing the effect of ELIXIR on the depolarization block threshold of control and ELIXIR-expressing neurons from WT Na$_V$1.6 (control $n = 8$ cells from 3 mice; ELIXIR $n = 13$ cells from 3 mice) and Na$_V$1.6 N1768D (control $n = 8$ cells from 2 mice; ELIXIR $n = 7$ cells from 2 mice) mice. Each bar, mean ± SEM. *$p < 0.01884$ (WT Na$_V$1.6 control), *$p = 0.0289$ (Na$_V$1.6 N1768D ELIXIR) compared with Na$_V$1.6 N1768D control by a two-way ANOVA followed by a Tukey's multiple comparisons test.

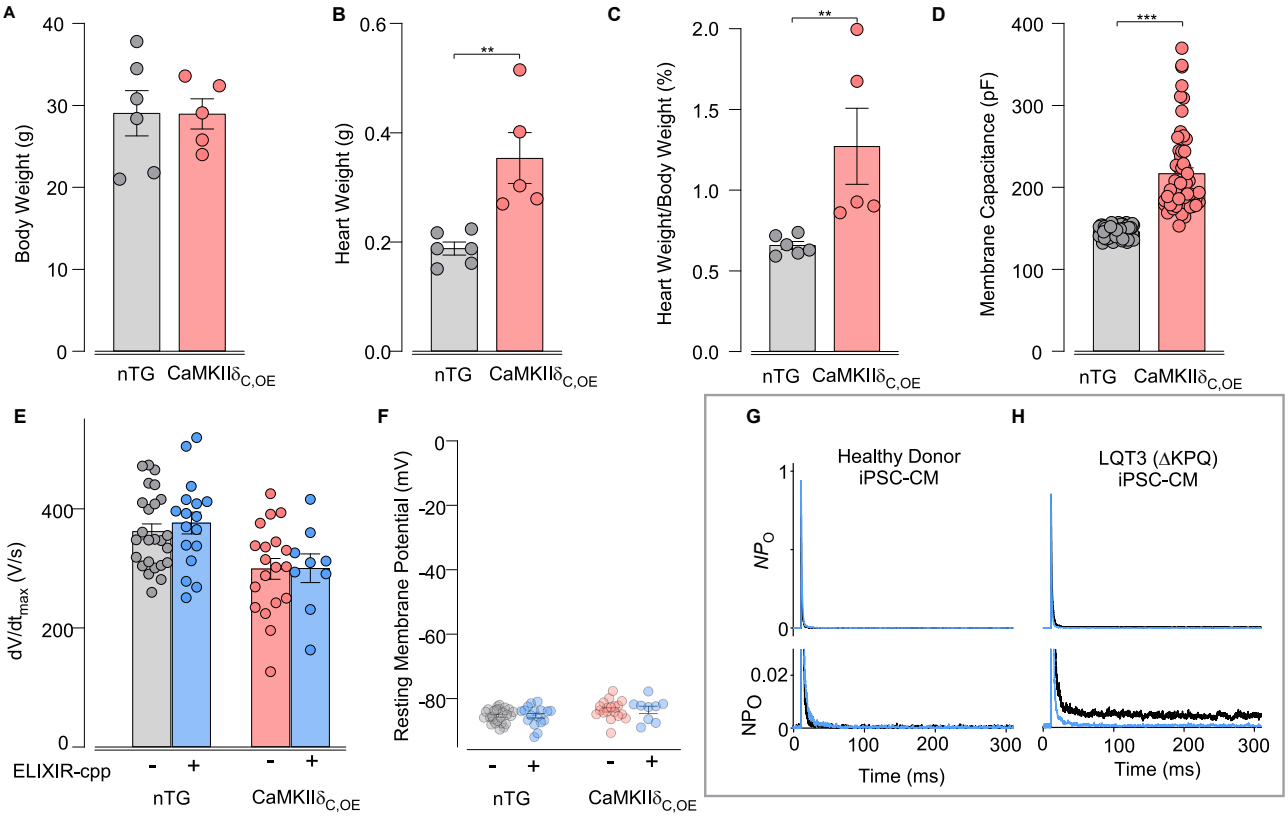


Figure S8. Comparison of nTG and CaMKII$\delta_{C,OE}$ murine heart morphology and extended analysis of hiPSC-CM, related to Figures 6 and 7

(A–C) Bar graph comparison of the body weight (A), heart weight (B), and heart weight to body weight ratio (C) of nTG (n = 6 mice) and CaMKII$\delta_{C,OE}$ (n = 5 mice) mice. Data are shown as mean ± SEM. $^{**}p$ = 0.004 for both heart weight and heart weight/body weight ratio by a Mann-Whitney U test.

(D) Bar graph comparison of the membrane capacitance of cardiomyocytes isolated from nTG (n = 80 cells from 6 mice) and CaMKII$\delta_{C,OE}$ (red, n = 56 cells from 5 mice) mice. $^{***}p < 0.001$ by a Mann-Whitney U test.

(E and F) Comparison of the dV/dt (E) and resting membrane potential (F) of cardiomyocytes isolated from nTG mice in the absence (n = 25 cells from 6 mice) and presence (n = 17 cells from 6 mice) of ELIXIR-cpp or from CaMKII$\delta_{C,OE}$ mice in the absence (n = 19 cells from 5 mice) or presence (n = 9 cells from 4 mice) of ELIXIR-cpp.

(G and H) Comparison of the ensemble average NP_o of HD (G), LQT3 (ΔKPQ) (H) iPSC-CMs, infected with adenovirus containing GFP (black) or ELIXIR (light blue). The lower panel shows an expanded view near the baseline.